

Interpretable Factors of Firm Characteristics *

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Abstract

We develop a new framework for constructing factors from firm characteristics that balances statistical efficiency and economic interpretability. Instead of using all characteristics equally, our method groups related characteristics and extracts one factor per group. Applied to the IPCA model by [Kelly, Pruitt and Su \(2019\)](#), this approach produces economically meaningful factors that match or exceed standard IPCA in pricing performance. Empirically, we show our parsimonious and transparent factors offer new insights into the drivers of stock returns and outperform existing benchmarks in out-of-sample tests, demonstrating the value of embedding economic structure into statistical modeling. Our framework reveals a clean separation between risk-based factors and those capturing persistent mispricing, offering a sharper lens into the economic channels behind return predictability.

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1 Introduction

A central challenge in empirical asset pricing is understanding how firm characteristics relate to expected returns. A large literature documents that hundreds of firm-level signals are correlated with cross-sectional variation in average returns (e.g., [Harvey et al. \(2016\)](#); [Hou et al. \(2020\)](#); [Chen and Zimmermann \(2022\)](#)). However, many of these characteristics are highly collinear, which complicates the identification of distinct priced factors and obscures the economic interpretation of risk premia. As [Hou et al. \(2020\)](#) emphasize, this high-dimensional setting often leads to redundant or overlapping information. In his influential AFA presidential address, [Cochrane \(2011\)](#) posed two enduring questions: How many factors are needed to explain expected returns? And can additional factors provide new economic insight or interpretation beyond standard models such as the five-factor model of [Fama and French \(2015\)](#)?

Two broad approaches have emerged to address these questions. The first builds on principal component analysis (PCA). Early studies such as [Connor and Korajczyk \(1986\)](#) extract latent factors from returns, and more recent work extends PCA to firm characteristics. However, PCA-based factors are typically dense linear combinations of all characteristics, making them difficult to interpret economically. For instance, if the dataset contains both value- and growth-related signals, any PCA factor may load on both, diluting the economic meaning. Moreover, traditional PCA assumes static loadings, limiting its ability to capture time-varying exposures. [Kelly et al. \(2019\)](#) propose Instrumented PCA (IPCA), which models dynamic loadings as functions of observable firm characteristics. While IPCA improves predictive performance, its estimated factors often remain diffuse and lack dominant economic themes, hindering economic interpretation and transparency.

A second approach leverages recent advances in machine learning (ML). A growing literature (e.g., [Gu et al. \(2020\)](#); [Feng et al. \(2020\)](#); [Freyberger et al. \(2020\)](#); [Kozak et al. \(2020\)](#))

applies ML tools to identify return-predictive characteristics. These methods handle nonlinearity and high dimensionality effectively, but often struggle with collinearity and identify overlapping signals. As a result, ML-based approaches may detect a large number of statistically significant signals that reflect overlapping information, limiting their applications in constructing parsimonious and interpretable factor models to facilitate economic understanding.

This paper proposes a simple and effective two-step framework to address both dimensionality and interpretability challenges in characteristic-based asset pricing. First, we partition the set of firm characteristics into economically meaningful and statistically coherent clusters. This structure can be guided by domain knowledge or inferred directly from the data. Second, we extract one latent factor per cluster that captures the dominant economic signal driving variation within the group.

This approach has two key advantages. First, by grouping characteristics that reflect the same economic mechanism, each resulting factor is tied to a clear risk source and becomes economically interpretable. Second, by avoiding redundant signals, the method yields a more parsimonious factor structure, reducing noise while preserving pricing power. The clustering assumption reflects a natural extension of the standard factor model: if asset returns are driven by a low-dimensional structure, then characteristics proxying for the same factor should be statistically related. In this sense, our clustering assumption is no stronger than the standard factor structure assumption. Specifically, we posit that characteristics within a cluster share a common economic origin or represent various noisy measurements of the same latent factor.

We implement this methodology by extending the IPCA model of [Kelly et al. \(2019\)](#) into a *Cluster-IPCA (C-IPCA)* framework. In C-IPCA, each factor loading is restricted to depend only on the characteristics within its corresponding cluster. As a result, each factor represents a specific economic theme, and each firm's exposure to that factor is estimated transparently from relevant characteristics within the corresponding cluster.

Imposing this cluster structure introduces a bias–variance tradeoff: while restrictions may introduce bias if clusters are misspecified, they can reduce variance and improve robustness. Ultimately, the tradeoff is empirical. Our results show that C-IPCA delivers both interpretable factors and strong out-of-sample performance.

Empirically, the C-IPCA model exhibits two desirable properties. First, each estimated factor is clearly related to a different source of economic risk, significantly improving both transparency and interpretability. Second, it provides new insights on what drives the cross-section of stock returns. The top four interpretable factors, based on Sharpe ratios, are: Operating Illiquidity (OI), Return Volatility (RV), Operating Efficiency (OE), and Size & Illiquidity (S&I). While the fourth one, the S&I factor, is a key component in several classical factor models, such as the SMB factor (Small Minus Big) in the Fama-French 3-factor model ([Fama and French, 1993](#)), the q-factor model ([Hou et al., 2015](#)), and the Fama-French 5-factor model ([Fama and French, 2015](#)), the other three factors underscore the importance of information asymmetry, financing frictions, and trading frictions in the stock market, as highlighted in previous studies ([Sloan, 1996](#); [Palazzo, 2012](#); [Grullon et al., 2012](#)). They collectively emphasize the important role of imperfect arbitrage that contributes to new factors beyond the CAPM. If there were frictionless arbitrage everywhere, all mispricing would disappear.

At the same time, some C-IPCA factors—e.g., Price Delay (PD), Investment (Inv), and Value (Val)—have low Sharpe ratios and limited economic significance. This motivates factor selection strategies. Using ordered selection and Bayesian model selection ([Chib et al., 2020](#)), we construct tangency portfolios from subsets of C-IPCA factors and find that they often outperform those built from full or selected IPCA factors.

Interestingly, we find that traditional asset pricing models - including the Fama-French three-factor model [Fama and French \(1993\)](#), the Fama-French five-factor model [Fama and French \(2015\)](#), and the q-factor model [Hou et al. \(2015\)](#) - are only able to explain a subset

of C-IPCA factors with low Sharpe ratios. In contrast, for the top-performing C-IPCA factors, these models produce significant alphas, underscoring that our approach captures novel pricing information beyond existing frameworks.

We further investigate which factors drive the alphas, the pricing errors in a factor model, which is a question largely ignored in the existing literature. To address this gap, we extend the C-IPCA framework to jointly capture risk-based and mispricing-based components by applying a group-lasso penalty to the estimation. Empirically, we find that factors such as Operating Illiquidity (OI), Operating Efficiency (OE), and the Market (Mkt) factor exhibit high Sharpe ratios with negligible alphas, indicating that their pricing power primarily reflects systematic risk premia. In contrast, Momentum, Profitability, Growth, and Intangibles explain a substantial share of alpha variation but display lower Sharpe ratios, consistent with persistent mispricing. This extension reveals a fundamental insight: the framework of interpretable factors naturally distinguishes between those representing systematic risk compensation and those capturing economically meaningful anomalies.

Furthermore, we provide evidence on the mechanism underlying the comparable performance (even outperformance) of the C-IPCA and IPCA models. In principle, the performance may arise from three sources: domain knowledge, the clustering information implied by the data and the reduction in parameter estimation resulting from the model structure. We find that both domain knowledge and data-driven clustering contribute to its success, while parameter parsimony plays a minor role. Our findings emphasize that embedding structure informed by economic theory and statistical regularities improves both interpretability and pricing power.

Our paper contributes to three strands of the literature: (i) dimensionality reduction and factor construction in asset pricing, (ii) interpretability and transparency in ML-based models, and (iii) the distinction between risk premia and mispricing. While related to [Stambaugh and Yuan \(2017\)](#), who apply clustering to time-series factors, our approach focuses on the cross-section,

handles large-scale data, and extracts latent cluster-level factors rather than simple averages.

The remainder of the paper is organized as follows. Section 2 describes the construction of factor models based on cluster analysis. Section 3 introduces the data and clustering methodology. Section 4 presents the clustering result and evaluates its effectiveness. Section 5 presents empirical results on the performance of various factor models. Section 6 investigates mechanism underlying the performance advantage of the C-IPCA models and presents some robustness tests. Section 7 concludes.

2 Model

This section describes the model which extends the IPCA approach of Kelly et al. (2019) by incorporating characteristic-based clustering to improve interpretability. Our method, Cluster-IPCA (C-IPCA), integrates economic structure into the IPCA by restricting factor loadings to depend only on characteristics within statistically and economically coherent clusters. We first briefly review the standard IPCA model and then detail how cluster-based restrictions can be incorporated to yield interpretable and economically grounded factors.

2.1 IPCA model

The IPCA model, proposed by Kelly et al. (2019), is a conditional factor model that allows for time-varying risk exposures by using firm characteristics as instruments. The key element of IPCA is that it models factor loadings (i.e., risk exposures) as linear functions of observable firm characteristics.

Specifically, the model consists of two equations. The return equation is:

$$r_t = \beta_{t-1} f_t + e_t, \quad (1)$$

and the exposure equation is:

$$\boldsymbol{\beta}_{t-1} = \mathbf{X}_{t-1}\boldsymbol{\Gamma} + \mathbf{u}_{t-1}, \quad (2)$$

where $\mathbf{r}_t = (r_{1t}, \dots, r_{Nt}) \in \mathbb{R}^N$ is the vector of excess returns for N assets at time t , $\mathbf{f}_t \in \mathbb{R}^J$ is a vector of J latent factors, $\boldsymbol{\beta}_{t-1} \in \mathbb{R}^{N \times J}$ contains the factor loadings (risk exposures) for each asset, $\mathbf{X}_{t-1} = (\mathbf{x}_{1,t-1}, \mathbf{x}_{2,t-1}, \dots, \mathbf{x}_{I,t-1}, 1)$ is an $N \times (I+1)$ matrix containing I firm characteristics and a constant term. $\boldsymbol{\Gamma} \in \mathbb{R}^{(I+1) \times J}$ contains the characteristic loadings that map characteristics to risk exposures.

For simplicity and following the empirical implementation in [Kelly et al. \(2019\)](#), we assume that the pricing errors (alphas) are zero. This restriction allows for more tractable estimation and is standard in the literature applying IPCA. The IPCA model can be estimated by minimizing the value-weighted mean squared error (MSE) of the pricing residuals,

$$\min_{\boldsymbol{\Gamma}, \{\mathbf{f}_t\}_{t=1}^T} \sum_{t=1}^T (\mathbf{r}_t - \mathbf{X}_{t-1}\boldsymbol{\Gamma}\mathbf{f}_t)' \mathbf{W}_{t-1} (\mathbf{r}_t - \mathbf{X}_{t-1}\boldsymbol{\Gamma}\mathbf{f}_t), \quad (3)$$

where $\mathbf{W}_{t-1}$ is the weighting matrix at time $t-1$. Following [Kelly et al. \(2019\)](#), we use a diagonal value-weighted matrix for $\mathbf{W}_{t-1}$. Different from [Kelly et al. \(2019\)](#), who employ equal weighting, we use value-weighted portfolios to assign greater importance to larger firms and mitigate concerns that results are driven by micro-cap stocks, as pointed out by [Hou et al. \(2020\)](#). This approach better reflects the economic significance of larger firms and aligns with practical considerations in portfolio management. Our results remain robust when equal-weighted portfolios are used, as reported in Appendix B, confirming that the findings are not sensitive to the weighting scheme.

The first-order conditions for this optimization problem yield the following recursive updat-

ing rules. The estimated factors are given by:

$$\hat{\mathbf{f}}_t = \left(\hat{\mathbf{\Gamma}}' \mathbf{X}_{t-1}' \mathbf{W}_{t-1} \mathbf{X}_{t-1} \hat{\mathbf{\Gamma}} \right)^{-1} \hat{\mathbf{\Gamma}}' \mathbf{X}_{t-1}' \mathbf{W}_{t-1} \mathbf{r}_t, \quad (4)$$

and the estimated characteristic loadings $\hat{\mathbf{\Gamma}}$ satisfy:

$$\text{vec}(\hat{\mathbf{\Gamma}}') = \left(\sum_{t=2}^T \mathbf{X}_{t-1}' \mathbf{W}_{t-1} \mathbf{X}_{t-1} \otimes \hat{\mathbf{f}}_t \hat{\mathbf{f}}_t' \right)^{-1} \left(\sum_{t=2}^T \left[\mathbf{X}_{t-1}' \mathbf{W}_{t-1} \otimes \hat{\mathbf{f}}_t' \right]' \mathbf{r}_t \right). \quad (5)$$

These equations are solved recursively using the algorithm by [Kelly et al. \(2019\)](#). Importantly, Equation (4) reveals that the estimated factors $\hat{\mathbf{f}}_t$ can be interpreted as portfolio returns, where the weights are given by:

$$\left(\hat{\mathbf{\Gamma}}' \mathbf{X}_{t-1}' \mathbf{W}_{t-1} \mathbf{X}_{t-1} \hat{\mathbf{\Gamma}} \right)^{-1} \hat{\mathbf{\Gamma}}' \mathbf{X}_{t-1}' \mathbf{W}_{t-1},$$

which is available at period $t - 1$. [Kelly et al. \(2019\)](#) also provide a formal econometric theory to guarantee the consistency and convergence of the IPCA estimators.

2.2 C-IPCA model

We now describe how to incorporate clustering into the IPCA framework to construct the Cluster-IPCA (C-IPCA) model. The standard IPCA model assumes that each risk exposure (i.e., factor loading) is a linear function of *all* firm characteristics, as specified in Equation (2). In contrast, C-IPCA imposes economic structure by assuming that each risk exposure depends only on a subset of firm characteristics —specifically, those within a single cluster representing a shared economic interpretation.

Formally, let $\{P_k\}_{k=1}^K$ denote the set of K clusters of firm characteristics, where each P_k indexes the characteristics grouped into the k^{th} cluster. The C-IPCA model introduces two

restrictions on Equation (2):

Restriction 1 (Cluster-Based Loadings). The k^{th} factor loading is a linear function only of the firm characteristics in the k^{th} cluster. That is, for each $k = 1, \dots, K$, we impose:

$$\Gamma_{ik} = 0, \quad \forall i \notin P_k, \quad (6)$$

where Γ_{ik} denotes the i^{th} row and k^{th} column of the loading matrix $\mathbf{\Gamma}$ in Equation (2), which reflects the contribution of the i^{th} firm characteristic to the k^{th} factor exposure. This restriction ensures that the k^{th} factor exposure and the corresponding factor are constructed solely from the characteristics in cluster P_k , thereby enhancing interpretability by aligning each factor with a distinct set of economically related variables.

Restriction 2 (Zero-Correlation Factor). We introduce an additional $(K + 1)^{\text{th}}$ factor whose loadings on all firm characteristics are set to zero:

$$\Gamma_{i,K+1} = 0, \quad \forall i = 1, \dots, I. \quad (7)$$

With this restriction, this factor exposure is restricted to be not related to any characteristics. This factor is included to capture variation in returns that is not explained by observable characteristics. Specifically, it allows for the possibility of a factor whose exposure is orthogonal to all characteristics — such as the market factor — which may otherwise be omitted under the cluster-based structure. Note that such a factor is a special case of the IPCA formulation in which all exposures are constant across assets. In practice, this $(K + 1)^{\text{th}}$ factor — referred to as the *zero-correlation (ZC)* factor — is often highly correlated with the market; in our application, its correlation with the market factor exceeds 0.999 as shown in Table 1.

These two restrictions jointly define the C-IPCA model, which combines the statistical

strength of IPCA with an economically grounded structure based on characteristic clustering. The resulting model enhances interpretability without compromising pricing performance, as we demonstrate in the empirical analysis.

With the two restrictions, the C-IPCA model takes the following form:

$$\mathbf{r}_t = \boldsymbol{\beta}_{1,t-1}f_{1,t} + \boldsymbol{\beta}_{2,t-1}f_{2,t} + \cdots + \boldsymbol{\beta}_{K+1,t-1}f_{K+1,t} + \mathbf{e}_t, \quad (8)$$

where the model includes at most $K + 1$ factors: one for each of the K characteristic clusters, and one zero-correlation (ZC) factor unrelated to any characteristics. In contrast, the standard IPCA model allows factor loadings to depend on all characteristics, potentially leading to a much larger number of latent factors.

Importantly, not all clusters necessarily contribute significantly to the model. In practice, statistical tests can be employed to eliminate uninformative clusters, resulting in a model with fewer than $K + 1$ factors, which will be discussed in the section on model selection 5.4.

An illustrative example. To better understand the role of the two restrictions, consider a simple illustrative example involving four firm characteristics: two momentum-related variables, mom_1 and mom_2 , and two value-related variables, value_1 and value_2 . Suppose the true return-generating process is driven by three latent factors. Under the standard IPCA framework, the model takes the following form (suppressing asset and time subscripts for simplicity):

$$r = \beta_1 f_1 + \beta_2 f_2 + \beta_3 f_3 + e, \quad (9)$$

where r is the return of a stock, f_j is the j^{th} factor, and β_j is the corresponding factor exposure. Each factor exposure β_j is further modeled as a linear combination of all characteristics,

regardless of their economic grouping,

$$\beta_j = \gamma_{1j} \cdot \text{mom}_1 + \gamma_{2j} \cdot \text{mom}_2 + \gamma_{3j} \cdot \text{value}_1 + \gamma_{4j} \cdot \text{value}_2 + \gamma_{5j} + u_j, \quad j = 1, 2, 3. \quad (10)$$

In contrast, the C-IPCA model imposes structure by restricting each exposure to depend only on the relevant cluster. Specifically, it assumes:

$$\beta_1 = \gamma_{11} \cdot \text{mom}_1 + \gamma_{21} \cdot \text{mom}_2 + \gamma_{51} + u_1, \quad (11)$$

$$\beta_2 = \gamma_{32} \cdot \text{value}_1 + \gamma_{42} \cdot \text{value}_2 + \gamma_{52} + u_2, \quad (12)$$

$$\beta_3 = \gamma_{53} + u_3. \quad (13)$$

Thus, in the C-IPCA framework, each factor exposure is tied to a single cluster of characteristics. For example, β_1 depends only on the momentum characteristics mom_1 and mom_2 , while β_2 depends only on the value characteristics value_1 and value_2 . The third factor loading β_3 is constant and unrelated to any characteristics, capturing the contribution of a latent factor — such as the market — that is not spanned by observable firm-level characteristics. This example illustrates how C-IPCA achieves a cleaner, economically interpretable structure while still accommodating latent variation uncorrelated with known signals.

Based on this example, we demonstrate how the two restrictions shape the structure of the model. First, Restriction 1 in Equation (6) directly leads to Equations (11) and (12). Specifically, to derive these two equations from the general specification in Equation (10), we restrict each factor exposure to depend only on the characteristics in a single cluster. Formally, we impose:

$$\begin{aligned} \gamma_{i1} &= 0, & \text{if } i \notin P_{\text{mom}}, \\ \gamma_{i2} &= 0, & \text{if } i \notin P_{\text{value}}, \end{aligned} \quad (14)$$

where $P_{\text{mom}} = \{1, 2\}$ denotes the momentum cluster and $P_{\text{value}} = \{3, 4\}$ denotes the value cluster.

ter. This constraint corresponds to the cluster-based restriction defined in Equation (6).

Second, Restriction 2 in Equation (7) directly lead to Equations (13). Specifically, to derive the Equation (13) from the general specification in Equation (10), we assume that one risk exposure is unrelated to any firm characteristics. This implies:

$$\gamma_{i3} = 0, \quad \text{for } i = 1, 2, 3, 4, \quad (15)$$

which corresponds to the zero-correlation (ZC) restriction in Equation (7).

Together, these restrictions yield the structured and interpretable form of the C-IPCA model, aligning each factor with a distinct economic theme while allowing for unobservable but priced sources of risk.

Estimation of the C-IPCA model. The C-IPCA model can be estimated using the same procedure as the standard IPCA model, since the objective function remains unchanged (see Equation (3)). The only difference lies in the additional linear restrictions imposed on the loading matrix, as specified in Equations (6) and (7). These restrictions modify only the first-order conditions for estimating the loadings. The equation for estimating the latent factors (Equation (4)) remains unchanged, as the constraints apply exclusively to the loading matrix $\mathbf{\Gamma}$. In contrast, the estimation of $\mathbf{\Gamma}$ (Equation (5)) is altered by omitting first-order conditions corresponding to parameters constrained to zero.

Specifically, Equation (5) includes $(I + 1) \times J$ first-order conditions—one for each element of $\mathbf{\Gamma}$. The $((i - 1) \cdot J + j)^{\text{th}}$ condition corresponds to the derivative with respect to Γ_{ij} . Under the C-IPCA restrictions, we drop all conditions associated with Γ_{ij} terms set to zero by either the cluster-based restriction (6) or the zero-correlation restriction (7). The remaining unrestricted parameters are then estimated from the reduced set of first-order conditions.

We solve this modified system recursively using an algorithm analogous to that in [Kelly et al.](#)

(2019). The resulting estimator retains the analytical tractability of IPCA while introducing structure that enhances interpretability.¹

2.3 OOS estimation

We briefly describe the out-of-sample (OOS) estimates for both the IPCA and C-IPCA models. These OOS estimates form the basis for the model performance evaluation in the subsequent sections. Starting from month $t \geq 180$, we recursively estimate the parameter matrix $\hat{\Gamma}$ using all data through time t , following the procedures outlined in Section 2.1 for IPCA and Section 2.2 for C-IPCA. Given $\hat{\Gamma}$, we compute the one-step-ahead out-of-sample factor returns at time $t + 1$ as:

$$\hat{f}_{t+1} = \left(\hat{\Gamma}' \mathbf{X}_t' \mathbf{W}_t \mathbf{X}_t \hat{\Gamma} \right)^{-1} \hat{\Gamma}' \mathbf{X}_t' \mathbf{W}_t \mathbf{r}_{t+1}, \quad (16)$$

which corresponds to the return on a managed portfolio with weights:

$$\left(\hat{\Gamma}' \mathbf{X}_t' \mathbf{W}_t \mathbf{X}_t \hat{\Gamma} \right)^{-1} \hat{\Gamma}' \mathbf{X}_t' \mathbf{W}_t.$$

Because all parameters are estimated using data available only through period t , the computed factor returns at time $t + 1$ are genuinely out-of-sample. We repeat this recursive estimation procedure each period, thereby constructing a full time series of OOS factor returns for both models.

¹As with all latent factor models, additional assumptions are required for identification. In IPCA, the factor and loading matrices are only identified up to rotation: for any nonsingular $K \times K$ matrix R , $(\Gamma R^{-1}, R \mathbf{f}_{t+1})$ yield observationally equivalent solutions. Kelly et al. (2019) impose three normalization conditions to resolve this issue. Since C-IPCA imposes additional structure, we adopt two assumptions to ensure uniqueness: (i) the monthly standard deviation of $\hat{f}_t$ is fixed at 1%, and (ii) the mean of $\hat{f}_t$ is restricted to be non-negative, following the third condition in Kelly et al. (2019).

3 Clusters

This section describes the data used in our analysis and outlines the clustering procedures applied to firm characteristics. We consider two distinct approaches to constructing clusters. The first relies purely on domain knowledge from the asset pricing literature, grouping characteristics according to established economic themes such as value, momentum, profitability, and investment. The second approach integrates both domain knowledge and data-driven insights, leveraging empirical patterns in the data to refine or reallocate characteristics within economically meaningful groups. This hybrid method aims to balance theoretical interpretability with improved empirical performance.

3.1 Data

We use the 94 firm characteristics introduced by [Gu et al. \(2020\)](#) for the U.S. equity market, covering the period from January 1985 to December 2021.² Each characteristic is standardized cross-sectionally in each month by subtracting its mean and dividing by its standard deviation. Monthly stock return data are obtained from the CRSP database. We follow the data cleaning procedures described in [Gu et al. \(2020\)](#).

To ensure that all model evaluations are strictly out-of-sample, we divide the full sample into two sub-periods. The first 15 years (1985:01–1999:12) serve as the training sample for initial parameter estimation. The remaining 22 years (2000:01–2021:12) form the testing sample for evaluating out-of-sample (OOS) performance.³ At the end of each month in the testing period, model parameters are re-estimated using all data available through that month, ensuring that factor construction relies solely on information known at the time.

²The data are publicly available at <https://dachxiu.chicagobooth.edu>, and detailed descriptions of the characteristics are provided in Table A.6 of [Gu et al. \(2020\)](#).

³As a robustness check, we also follow [Kelly et al. \(2019\)](#) by using the first 10 years as the training period; results remain qualitatively unchanged.

3.2 Intuitive clusters (IC)

A common approach to organizing firm characteristics is to rely on domain knowledge and established economic theory. We refer to this as the *Intuitive Clustering* (IC) method. Under this approach, characteristics are grouped based on their underlying economic concepts — such as momentum, value, or profitability — rather than statistical similarity. (see [Hou et al., 2015](#), [Harvey et al., 2016](#), [McLean and Pontiff, 2016](#), [Hou et al., 2020](#), [Freyberger et al., 2020](#), and [Han et al., 2024](#).) The IC approach has been widely applied across various datasets, including the 202 characteristics in [Harvey et al. \(2016\)](#), the 452 characteristics in [Hou et al. \(2020\)](#), and the 299 characteristics in [Han et al. \(2024\)](#).

We apply a similar methodology to the 94 firm characteristics in our dataset. First, following [Hou et al. \(2015\)](#), [Hou et al. \(2020\)](#), [Freyberger et al. \(2020\)](#), and [Han et al. \(2024\)](#), we classify characteristics into six economically motivated groups (with abbreviations in parentheses): Momentum (Mom), Value (Val), Profitability (Prof), Investment (Inv), Intangibles (Int), and Trading Frictions (TFs).

Second, each firm characteristic is assigned to the cluster most closely aligned with its economic meaning. This classification yields six intuitive clusters containing 9, 8, 13, 12, 36, and 16 characteristics, respectively. These IC clusters serve as a benchmark for evaluating the interpretability and empirical performance of our factor models. Table [A1](#) shows the mapping between the ICs and firm characteristics.

3.3 Data-Driven clusters

While intuitive clusters (IC) are grounded in economic reasoning, they may overlook important statistical relationships among firm characteristics—relationships that can enhance both understanding and interpretation. This section introduces the *Data-Driven Cluster* (DC) ap-

proach, which aims to combine economic intuition with empirical information to form clusters that are both interpretable and statistically coherent.

How can we integrate economic priors with empirical signals in a principled manner? The Bayesian paradigm provides a natural framework. Under Bayesian inference, one begins with a prior belief—grounded in theory or existing knowledge—and updates it with data to obtain a posterior belief. Analogously, the DC method treats the intuitive clusters as prior groupings and refines them using observed statistical relationships among characteristics.

Within this framework, the following subsections describe how we integrate prior economic classifications with empirical information to construct data-driven clusters. Specifically, we proceed in two steps. First, in Section 3.3.1, we quantify the statistical similarity between characteristics based on their historical behavior. Second, in Section 3.3.2, we implement an agglomerative hierarchical clustering procedure that combines economic priors with empirical similarity structure to produce clusters that are both economically meaningful and statistically grounded.

3.3.1 Characteristic similarity

The similarity between characteristics i and j is defined as:

$$s_{ij} = \exp \left(- \left(1 - |\rho_{ij}| \right) \right), \quad (17)$$

where ρ_{ij} is the time-series average of the monthly, value-weighted, cross-sectional rank correlations:

$$\rho_{ij} = \frac{1}{T} \sum_{t=1}^T \frac{\sum_{n=1}^{N_t} w_t^n (x_{it}^n - \bar{x}_{it}) (x_{jt}^n - \bar{x}_{jt})}{\sqrt{\sum_{n=1}^{N_t} w_t^n (x_{it}^n - \bar{x}_{it})^2} \sqrt{\sum_{n=1}^{N_t} w_t^n (x_{jt}^n - \bar{x}_{jt})^2}}, \quad (18)$$

where x_{it}^n denotes the cross-sectional rank of characteristic i for stock n at time t , $\bar{x}_{it} = \sum_{n=1}^{N_t} w_t^n x_{it}^n$ is the value-weighted average rank for characteristic i at time t , w_t^n is the value weight of stock

n at time t , normalized such that $\sum_{n=1}^{N_t} w_t^n = 1$, N_t is the number of stocks at time t . In the Appendix B, we discuss the results for the equal weights.

The use of value-weighted rank correlations in Equation (18) is motivated by two considerations. First, using ranks helps mitigate the influence of outliers, ensuring that the correlation reflects the underlying economic relationships rather than extreme values. Second, by applying value-weighted ranks, we place greater emphasis on larger firms, consistent with our focus on value-weighted Sharpe ratios as discussed later. This approach aligns with Hou et al. (2020), which argues that anomalies are predominantly driven by micro- and small-cap stocks, which deliver limited information on tangency portfolio in practical portfolio management. By emphasizing value-weighted correlations, we mitigate the concern that our results may be disproportionately influenced by these smaller stocks, whose characteristics may not fully represent broader market trends.

There are several important considerations regarding the transformation in Equation (17), which converts rank correlations into pairwise similarities:

1. **Noise Robustness:** The exponential transformation, as used in Saxena et al. (2017) and Von Luxburg (2007), further down-weights the similarity for characteristic pairs with low rank correlations. This reduces the influence of noisy or weak correlations that might distort clustering outcomes, ensuring that only the most statistically significant relationships dominate the similarity measure.
2. **Correlation Strength:** Following Stambaugh and Yuan (2017), we interpret large magnitude of correlations as indicative of strong statistical relationships. This ensures that characteristics with high similarity are meaningful and robust, capturing relationships that truly contribute to the clustering process.
3. **Sign Invariance:** By taking the absolute value of the correlations, we ensure that neg-

atively but strongly correlated characteristics (e.g., size and illiquidity, with $\rho = -0.87$) are treated as similar. Despite their opposite signs, these characteristics likely reflect the same underlying economic factor and thus should be grouped together.

We will use the similarity metric to refine the initial IC clusters in the subsequent steps of the Data-Driven Cluster (DC) method. The DC method adjusts the clusters by integrating both the economic intuition of the IC method and the statistical insights captured by the similarity structure between characteristics.

3.3.2 Posterior adjustment process

The DC method refines the initial intuitive cluster by integrating prior economic knowledge with data. To achieve this, we adopt the framework proposed by [Karypis et al. \(1999\)](#), utilizing a split-and-merge approach to adjust the original clustering (hereafter, we refer the algorithm to as Chameleon).

The core idea of this approach is to begin with the initial IC clusters, then iteratively partition them into smaller sub-clusters, and finally merge those sub-clusters to get final clusters. As for partition process, its goal is to minimize the similarity between clusters while maximizing the similarity within each cluster. This process is controlled by a hyperparameter, denoted as the number of sub-clusters, m . A key advantage of this process is that the resulting sub-clusters retain interpretability, as they are derived from the initial intuitive clusters. Besides, as for merging process, it allows for the potential combination of sub-clusters from different initial intuitive clusters, resulting in more flexible and potentially more meaningful groupings. This process is governed by a hyperparameter, denoted as the number of final clusters, K . Overall, the iterative nature of this procedure allows the method to adapt to both the economic structure and the statistical patterns uncovered by the data.

Before introducing the specific steps of the clustering procedure, we first formalize the prob-

lem in a graph-based framework, following Guha et al. (1998) and Karypis et al. (1999). In this framework, firm characteristics are treated as vertices in a graph, and the edges between them capture the pairwise similarity structure among characteristics. The weight of edges reflect similarity: higher similarity corresponds to a larger weight and a shorter distance between vertices. For simplicity, we use “firm characteristics” and “vertices” interchangeably, and treat “low similarity,” “small edge weights,” and “large distances” as equivalent expressions throughout the discussion.

Then we introduce the specific steps of the clustering procedure. Following the Chameleon clustering method (Karypis et al., 1999), we perform a split-and-merge adjustment to the initial IC clusters through a three-step procedure.

Step 1: Graph Construction. The first step involves constructing a sparse graph based on the similarity between firm characteristics. In this process, edges are retained only between characteristics that are among the knn -nearest neighbors, where knn is a hyperparameter. This approach not only improves computational efficiency but also ensures that the most significant relationships between characteristics are preserved. Additionally, this step effectively diminishes the impact of small pairwise similarities by shrinking them towards zero. This is beneficial because small similarities could arise from noise in the data, which may distort the clustering process.

Step 2: Splitting IC Clusters. In this step, we partition the initial IC clusters into smaller sub-clusters. The number of sub-clusters, denoted as m , is a hyperparameter, which can be fine-tuned during the process. The partitioning is performed iteratively: in each iteration, the largest sub-cluster is selected and split into two smaller sub-clusters. The goal of the splitting process is to minimize the inter-cluster similarity, which is defined as the average similarity between characteristics in the two resulting sub-clusters. By doing so, we ensure that the resulting sub-clusters are as distinct as possible. This iterative process continues until exactly m sub-clusters

are obtained, which we refer to as basic sub-clusters.⁴

Step 3: Merging Sub-clusters. Starting with the m basic sub-clusters, we sequentially merge the two most similar sub-clusters. The merging criterion is based on the Relative Inter-Cluster Similarity (RIS), which is defined as:

$$\text{RIS}(C_i, C_j) = \frac{\text{INTER}(C_i, C_j)}{\frac{|C_i|}{|C_i|+|C_j|}\text{INTRA}(C_i) + \frac{|C_j|}{|C_i|+|C_j|}\text{INTRA}(C_j)}, \quad (19)$$

where $\text{INTER}(C_i, C_j)$ is the inter-cluster similarity, measured as the average edge weight between the two clusters, and $\text{INTRA}(C_i)$ and $\text{INTRA}(C_j)$ are the intra-cluster similarities of clusters C_i and C_j , respectively. At each merging step, the algorithm combines the two clusters with the highest RIS value. In other words, the algorithm merges clusters with the high inter-cluster similarity but low intra-cluster similarity. This ensures that the final clusters exhibit high internal coherence while minimizing external overlap.⁵

3.3.3 Advantages of the Data-Driven cluster method

There are two advantages of our Data-Driven Cluster (DC) method: (1) robustness to data noise, making it well-suited for financial data with a low signal-to-noise ratio, and (2) relaxed assumptions about the data, allowing for clusters of varying shapes and sizes as emphasized by [Karypis et al. \(1999\)](#).

Robustness to Data Noise. The merging rule in Equation (19) enhances Chameleon’s robustness to data noise. As highlighted by [Karypis et al. \(1999\)](#), this rule is specifically designed to adapt to datasets where clusters exhibit heterogeneous densities. In other words, it accom-

⁴Instead of using the hMetis algorithm as in [Karypis et al. \(1999\)](#) for partitioning, we adopt spectral clustering, which is more computationally efficient and has publicly available source code.

⁵The merging criterion in [Karypis et al. \(1999\)](#) is designed to balance two objectives: robustness to noise and adaptiveness to cluster shape, quantified by the metrics RC and RI , respectively. Given the low signal-to-noise ratio commonly observed in financial data, we place infinite weight on robustness to noise, effectively disregarding cluster shape adaptability. As a result, our merging rule focuses solely on robustness to noise, and the RIS metric introduced in our paper is equivalent to the RC metric defined in [Karypis et al. \(1999\)](#).

modates structures in which some clusters consist of firm characteristics with high similarity (high-density clusters), while others exhibit lower similarity (low-density clusters).

This versatility is particularly valuable in asset pricing, where noise frequently reduces the density of economically meaningful clusters. To illustrate, recall that firm characteristics within the same cluster are assumed to be noisy proxies for a common latent risk exposure. In the absence of measurement error, each characteristic would perfectly capture the latent exposure, yielding a correlation of one across characteristics within the cluster and maximizing the similarity measure $s_{i,j}$ in Equation (17). However, as measurement error increases, observed characteristics deviate from the latent exposure, reducing correlations and thereby lowering both similarity and cluster density. Because the Chameleon algorithm maintains effectiveness even in the presence of low-density clusters, it is particularly well-suited for analyzing financial data subject to substantial noise.

Relaxed Assumptions on Data Structure. In the third step of the clustering process (Section 3.3.2), Chameleon’s strategy of merging sub-clusters, rather than individual firm characteristics, allows it to relax assumptions about the data.⁶

3.3.4 Selection of hyper-parameters

Our algorithm involves three hyperparameters: (1) the number of nearest neighbors knn in Step 1, (2) the number of sub-clusters m in the splitting process (Step 2), and (3) the final number of clusters K in the merging process (Step 3). This section explains how to select the

⁶Traditional algorithms such as K-means assume clusters to be roughly elliptical and of similar size, which limits their flexibility. These methods typically represent each cluster with a *single vertex* (the “cluster representative”) and compute clustering results based on the distance from each data point to the cluster representative, called *vertex-to-vertex distance*. While effective when clusters are spherical and of comparable sizes, this approach performs poorly when clusters are irregularly shaped or vary widely in size, as the vertex-to-vertex distance fails to capture the overall shape and size of the clusters.

In contrast, Chameleon computes the similarity between sub-clusters based on the *sub-cluster-to-sub-cluster* distances, as reflected in the $INTER(C_i, C_j)$ term of the merging rule in Equation (19). This allows the algorithm to adapt to a wider variety of cluster shapes and sizes, ensuring that the final clustering structure better reflects the true underlying data distribution.

optimal values for these hyperparameters using a grid search.

We first define a grid of candidate values for each hyperparameter: $knn = \{10, 15, \dots, 90\}$, $m = \{16, 19, 24, 31\}$, and $K = \{1, 2, \dots, 15\}$, where m is chosen such that, on average, there are 3, 4, 5, or 6 vertices in each basic sub-cluster after Step 2.

Selection of K . We begin by selecting the optimal value for K given m and knn . This corresponds to determining when to stop the merging process in Step 3 in Section 3.3.2. Following the approach of Barton et al. (2019), we terminate the merging process when the maximum relative inter-cluster similarity (Max_RIS) becomes abnormally low. The Max_RIS is defined as the highest RIS value among all cluster pairs, and is calculated as:

$$Max_RIS = \max_{i,j=1,\dots,K,i \neq j} RIS(C_i, C_j), \quad (20)$$

where $RIS(C_i, C_j)$ represents the relative inter-cluster similarity between clusters C_i and C_j as in Equation (19).

The intuition behind using Max_RIS is as follows: at each stage, the merged pair has an RIS equal to the Max_RIS , because the algorithm always merges the pair with the highest RIS . An abnormally low Max_RIS indicates that the merger is inappropriate, either because the inter-cluster similarity is very low or the intra-cluster similarity is unusually high, suggesting the clusters are not well-matched. Therefore, we should stop merging at this point and consider the current number of clusters as the optimal K .

To determine at which point the Max_RIS is abnormally low — i.e., identifying the stopping point for merging — we follow the approach of Barton et al. (2019), which detects a sharp drop in Max_RIS . In our merging procedure, we always combine the two clusters with the highest RIS , so the value of Max_RIS naturally tends to decrease as the number of clusters K decreases. However, when this decrease becomes disproportionately large, we take it as evidence that the

Max_RIS has become abnormally low, marking the point at which we stop merging.

Specifically, the approach to determine the optimal K involves two steps. First, it computes a baseline measure for detecting the abnormal low Max_RIS values. This baseline measure is defined as the average reciprocal of Max_RIS within the larger half of K 's range (i.e. the range $[m/2, m]$ as K ranges from 1 to m). We denote this value as $\overline{Max_RIS}^{-1}$. Second, we search in descending order for K within the rest range $[1, m/2-1]$. Specifically, we look for the first K where the reciprocal Max_RIS is at least f times greater than the $\overline{Max_RIS}^{-1}$. When found, we take the preceding K as the optimal value. If this condition is not fulfilled for any K , we iteratively relax the threshold by adjusting it from f to $\frac{f}{\eta^i}$ times of $\overline{Max_RIS}^{-1}$, where i is an iteration index that starts at 1 and increases by 1 until the condition is fulfilled. Here f is the initial scaling value and η controls the rate of decrease in the threshold. Following [Barton et al. \(2019\)](#), we set f to 10^3 . The value of η depends on the problem, and in this case, we use $\eta = 1.3$. Results are robust if $\eta = 1.2$ or $\eta = 1.4$.

Selection of m and knn . Finally, we select the optimal values for m and knn by evaluating the performance of the corresponding C-IPCA models and select the hyperparameter with highest Sharpe ratio of the training sample for the tangency portfolio. Specifically, we construct the tangency portfolio using all model factors following a purely out-of-sample procedure: for each time t , we estimate the mean and covariance matrix of factor returns using data available up to time t , and then track the realized return of the tangency portfolio in time $t + 1$. The tangency portfolio is initially constructed in December 1989. The optimal hyper-parameters are $m = 24, knn = 55, K = 12$.

4 Data-Driven clusters vs. intuitive clusters

This section provides a comparison of the Data-Driven Cluster (DC) and Intuitive Cluster (IC) methods to highlight the role of empirical data in refining clusters beyond prior domain knowledge.

4.1 Connections between DC and IC

Figure 1 shows the clustering results for firm characteristics based on both the DC and IC methods. In the figure, the vertices represent firm characteristics, with different colors indicating the clusters assigned by the IC method. The number of vertices is for visualization purposes only and does not reflect the actual number of firm characteristics in the dataset. The dashed lines in the figure represent the clustering results for each method. For the details on the mapping between characteristics and clusters, please see the table in Appendix A1.

Panel A shows the clustering results based on the IC method, while Panel B shows the results from the DC method. In our data, the optimal number of clusters for the DC method is 12, with the following abbreviations: Momentum (Mom), Return Volatility (RV), Size and Illiquidity (S&I), Turnover (TO), Price Delay (PD), Investment (Inv), Growth (Gr), Profitability (Prof), Operating Illiquidity (OI), Operating Efficiency (OE), Intangibles (Int), and Value (Val).

There are several key observations when comparing the clustering results between Panel A (IC) and Panel B (DC).

1. **Momentum Cluster (Mom):** The Mom cluster in the IC method is fully aligned with the DC method, with the same firm characteristics grouped together, demonstrating a complete overlap.

2. **Trading Frictions (TFs) Cluster:** The TFs cluster in the IC method is further divided into four distinct clusters in the DC method: RV, S&I, TO, and PD. The RV cluster includes

characteristics such as idiosyncratic return volatility and overall return volatility. The S&I cluster contains characteristics like size and Amihud illiquidity. The TO cluster includes characteristics such as share turnover and the volatility of share turnover. The PD cluster includes characteristics like abnormal earnings announcement volume and price delay.

3. **Intangibles Cluster (Int):** The Int cluster in the IC method is split into three distinct clusters in the DC method: OI, OE, and Int. OI includes characteristics such as the quick ratio and current ratio, OE contains characteristics like the accruals, sales-to-inventory ratio, and Int includes characteristics such as R&D investment.

4. **Investment, Profitability, and Value Clusters:** In the IC method, the characteristics related to Investment (Inv), Profitability (Prof), and Value (Val) are reclustered into four separate clusters in the DC method: Gr (growth), Inv (investment), Prof (profitability) and Val (value).

These observations highlight the flexibility of the DC method in refining the initial intuitive clusters by leveraging empirical data. Although the IC method relies solely on prior economic knowledge on the characteristics of the group, the DC method adapts these groupings by exploiting the underlying statistical relationships among the characteristics, yielding a more nuanced clustering structure.

4.2 Evaluating the effectiveness of DC clustering

A core goal of clustering is to group together firm characteristics that are highly similar while separating those that are dissimilar. This subsection evaluates whether our Data-Driven Clustering (DC) method effectively achieves this goal. Specifically, we assess whether characteristics grouped together under DC exhibit high pairwise similarity and whether characteristics placed in different clusters exhibit low similarity.

We approach this evaluation from two complementary perspectives. First, we directly examine the pairwise similarity between firm characteristics within and across clusters. Second,

we visualize the clustering structure by mapping characteristics into a low-dimensional space that preserves pairwise distances, revealing how well the clusters are separated geometrically.

Pair-wise similarity. The DC method incorporates empirical information via the similarity measure s_{ij} . If the clustering structure is informed by data, we expect within-cluster similarity to be high and between-cluster similarity to be low.

Distance-based visualization via MDS. To provide an alternative perspective, we transform the similarity matrix into a distance matrix using the following transformation:

$$d_{ij} = \frac{1}{s_{ij}} - 1, \quad (21)$$

where s_{ij} is the similarity between characteristics i and j as defined in Equation (18), with $d_{ij} \geq 0$ and $d_{ii} = 0$. This transformation preserves the inverse relationship between similarity and distance: higher similarity implies shorter distance.

To visualize the clustering structure implied by these distances in a two-dimensional space to enhance our intuition, we apply Multidimensional Scaling (MDS), a statistical technique that maps objects to points in a low-dimensional space such that pairwise distances are preserved as closely as possible (Borg, 2011). Specifically, given a set of N firm characteristics and their pairwise distances d_{ij} , MDS map each characteristic into one point in a two-dimensional space by solving:

$$\text{Stress} = \sqrt{\frac{\sum_{i < j} (d_{ij} - \hat{d}_{ij})^2}{\sum_{i < j} d_{ij}^2}}, \quad (22)$$

where $\hat{d}_{ij}$ is the Euclidean distance between the mapped points of characteristics i and j . The objective is to minimize the stress function, thereby ensuring that the projected distances in the two-dimensional space approximate the original distances.

Given the large number of firm characteristics, it is inherently difficult to preserve all pairwise distances accurately in two-dimensional space. To enhance interpretability, we present the results in four separate panels in Figures 2 and 3, with each panel corresponding to a subset of DC clusters.

Figure 2a presents the results for the four trading frictions clusters in the DC method: Return Volatility (RV), Turnover (TO), Size & Illiquidity (S&I), and Price Delay (PD), which together include 15 firm characteristics.

The left panel displays a heatmap of the pairwise similarity matrix among these 15 characteristics. Darker colors indicate higher similarity. To enhance interpretability, the characteristics are ordered by DC cluster membership, and red grid lines partition the matrix into 16 submatrices, corresponding to within- and between-cluster similarities. The visual contrast in shading clearly reveals that within-cluster similarities are substantially higher than between-cluster similarities, consistent with the goal of the DC method to group statistically similar characteristics.

The right panel of Figure 2a shows the two-dimensional embedding of these characteristics using MDS. Each point represents a characteristic, and points are colored according to their DC cluster membership. The plot demonstrates that characteristics within the same cluster tend to be located near one another, while those from different clusters are well-separated. This spatial pattern provides additional evidence that the DC clustering method effectively reduces between-cluster similarity while enhancing within-cluster coherence.

Figure 2b shows the similarity between firm characteristics in the three intangible clusters in DC: OI, OE and Int. From the heatmap on the left, we can see that darker grid cells appear more frequently in the diagonal matrices enclosed by the red lines, indicating that firm characteristics with higher similarity are likely to be in the same cluster. Besides, there is an interesting finding in firm characteristics in the third diagonal matrix enclosed by the red lines. Their similarity with each other is not very high, and their similarity to other characteristics is also low. This

reflects the advantage of the Chameleon clustering method: low similarity may be due to high noise, and these firm characteristics should belong to the same cluster rather than being treated as separate clusters. This is also evident from the scatter plot on the right: some yellow vertices are far from all other vertices, which may be due to high noise. Each of these vertices should belong to the same cluster as other yellow vertices rather than forming a separate cluster.

Figure 3a displays the similarity structure among firm characteristics in the DC Investment (Inv) and Growth (Gr) clusters. The right panel shows the two-dimensional embedding of firm characteristics based on their pairwise distances, with colors representing DC cluster membership and shapes indicating IC cluster assignments. Several key patterns emerge. First, consider the IC Investment cluster (represented by diamond-shaped points). The DC method splits this group into two distinct sub-clusters, shown as yellow and blue diamonds. These two groups are well-separated in the embedded space, indicating that they are statistically dissimilar despite sharing a common economic label under the IC scheme. The DC method, therefore, refines the initial classification by recognizing and separating these empirically distinct subgroups.

Second, the DC method merges firm characteristics from the IC Profitability, Intangibles, and Investment clusters. For instance, yellow circular (IC Profitability), triangle (IC Intangibles), and diamond (IC Investment) points are all positioned close to one another, suggesting that they capture a shared empirical signal. Accordingly, the DC algorithm groups them into a single cluster, highlighting its ability to uncover cross-cutting relationships not reflected in the IC taxonomy.

Finally, while some points within the same DC cluster appear dispersed - such as the yellow circular points on the far left and right — this dispersion reflects the influence of prior economic classifications. These points belong to the same IC cluster and share a common economic interpretation (e.g., profitability), even though their empirical similarity s_{ij} is low. The DC method incorporates this prior structure and retains them in the same cluster, demonstrating its

flexibility in balancing data-driven evidence with domain knowledge.

Figure 3b presents the similarity structure of firm characteristics within the DC Profitability (Prof) and Value (Val) clusters. The heatmap on the left shows that the diagonal blocks enclosed by red lines — representing within-cluster similarities — are distinctly darker than the off-diagonal blocks, indicating strong intra-cluster similarity and weak inter-cluster similarity. This visual evidence suggests that the DC method effectively groups together characteristics with high empirical similarity while separating those that are dissimilar.

The right panel shows the two-dimensional embedding of firm characteristics based on the pairwise distance matrix. As in earlier panels in Figure 3a, colors denote DC cluster membership, while shapes indicate IC classifications. The figure clearly shows that characteristics assigned to the DC Value and Profitability clusters form two well-separated groups in the embedded space. This result reinforces the core strength of the DC methodology: after incorporating data-driven similarity measures, firm characteristics with strong empirical affinity are more likely to be grouped together, even when their intuitive classification is ambiguous or overlapping. These findings are consistent with the patterns observed in Figures 2 and 3a, further validating the effectiveness of our approach.

5 Performance of the C-IPCA model

Given the characteristic clusters constructed in Section 3, we proceed to estimate the C-IPCA model, which imposes cluster-specific restrictions on the characteristic coefficients in the factor loadings, as described in Section 2.2. In this section, we evaluate the empirical performance of the C-IPCA model under different clustering schemes for firm characteristics. This comparison allows us to assess the impact of clustering methodology on both the interpretability and pricing performance of the resulting factor models.

5.1 Interpretable factors

Given the estimated characteristic coefficient matrix $\hat{\Gamma}$, recall that each latent factor in the C-IPCA model can be expressed as a linear combination of individual stock returns, with weights determined by the projection formula (16). This formulation enables us to construct factor-mimicking portfolios directly from the estimated model parameters. Each factor is interpretable by design, as it is extracted from a distinct cluster of firm characteristics.

Figure 4 illustrates the interpretability of C-IPCA factors through an example. The figure displays the absolute values of the $\hat{\Gamma}$ matrix from Equation (2) for an IPCA model with 13 factors, estimated using the full sample. Each column corresponds to a factor, while each row represents a characteristic. The red solid lines partition the 94 characteristics into 13 clusters based on the DC structure. The cell at row i and column j represents the absolute value of the loading of the i -th characteristic on the j -th factor exposure, reflecting the importance of that characteristic for the corresponding factor exposure. Darker shading indicates larger absolute loadings, signifying higher importance, as characteristics have been standardized to have zero mean and unit variance. To enhance visibility of within-column variation, each column is scaled so that the sum of squared elements equals 1.

Figure 4a demonstrates that in the standard IPCA model, the loadings for each factor are dispersed across multiple clusters, complicating factor interpretation. For example, in the second column, both market beta and momentum exhibit large loadings, making it difficult to classify the corresponding factor as primarily a market or momentum factor. In contrast, Panel (b) shows that under the C-IPCA model, loadings for each factor are highly concentrated within a single cluster of characteristics. For instance, the first column exhibits significant loadings exclusively on operating efficiency (OE)-related characteristics, indicating that the associated risk exposure is driven predominantly by OE. This clustering enables a clear and economically meaningful interpretation of each factor.

Table 1 shows the summary statistics on the properties of interpretable factors under both the IC-IPCA (intuitive clustering) and DC-IPCA (data-driven clustering) models. Panel A reports results for IC-IPCA, while Panel B presents the corresponding outcomes for DC-IPCA. Each row represents a latent factor, with the final column indicating the economic label or abbreviation of its associated characteristic cluster. To aid interpretation, factors are sorted by their out-of-sample Sharpe ratios over the testing period (2000:01–2021:12), except for the market factor, which is listed in the final row.

The second through fifth columns report standard performance metrics, including the time-series mean (Mean), standard deviation (SD), annualized Sharpe ratio (Sharpe), and maximum drawdown (MDD) of monthly factor returns. The Sharpe ratio is computed as the ratio of the time-series mean to the standard deviation, and annualized by multiplying the monthly Sharpe by $\sqrt{12}$. MDD is defined as the largest cumulative loss from peak to trough over the sample period and serves as an indicator of downside risk.

Panel A: Performance of IC-IPCA Factors. Among the seven factors estimated under IC-IPCA, the two with the highest Sharpe ratio are the Investment (Inv) and Intangibles (Int) factors, with annualized Sharpe ratios of 0.62 and 0.44, respectively. In terms of downside risk, the Momentum (Mom) factor exhibits the largest maximum drawdown, consistent with the findings of [Daniel and Moskowitz \(2016\)](#).

Panel B: Performance of DC-IPCA Factors. Results from DC-IPCA corroborate and refine the insights from Panel A. Among the thirteen factors, the top two performers are the Operating Illiquidity (OI) and Return Volatility (RV), with Sharpe ratios of 0.73 and 0.56, respectively. These results are consistent with Panel A's strong performance of the Intangibles factor, whose cluster includes OI. In terms of risk, the S&I and Mom factors have the highest drawdowns, mirroring the risk pattern in Panel A.

Several DC-IPCA factors exhibit relatively low Sharpe ratios — namely, Price Delay (PD),

Investment (Inv), and Value (Val) — with values below 0.15 (0.13, 0.08, and 0.02, respectively). This suggests that not all DC-IPCA factors are equally important for pricing cross-sectional returns, and a parsimonious factor model may retain only a subset of the constructed factors.

Panel C: Market Factor Correlation. Panel C reports the correlations between market factors (MktRf) and the following factors. (1) The zero-correlation factors (ZC) of IC-IPCA and DC-IPCA. (2) The IPCA factor most correlated with the market return among all factors in a IPCA model. For comparison, we estimate two versions of the standard IPCA model, one with seven factors (IPCA7) and another with 13 (IPCA13) that match the number of factors in the IC-IPCA and DC-IPCA models, respectively. We denote the IPCA factor most correlated with the market return as MF(IPCA7) or MF(IPCA13). The benchmark market factor (MktRf) is the value-weighted excess return on the aggregate stock market.

Across all models, we find that the estimated factors are highly correlated with the true market factor, with correlations near 1. This result confirms that the ZC factor in C-IPCA can be interpreted as a market-mimicking portfolio. It also supports the use of restriction (7) in the C-IPCA model. Although standard IPCA does not impose the existence of a market factor, one naturally emerges from the data. This reinforces the necessity of explicitly incorporating a market factor in the C-IPCA specification to ensure robust empirical performance and interpretability.

5.2 Comparison with traditional factors

A natural and interesting question is how our clustering-based factors relate to traditional asset pricing factors. To address this, Panels A and B of Table 1 report the alphas of our factors using Fama–French regressions. Specifically, columns six through eight present alphas from regressions of each factor on standard factor models, including the Fama–French three-factor model (FF3; Fama and French, 1993), the Fama–French five-factor model (FF5; Fama and

[French, 2015](#)), and the q-factor model (Q4; [Hou et al., 2015](#)).

Two main findings emerge. First, while traditional factors explain some of our clustering-based factors, this explanatory power is concentrated among factors with the lowest Sharpe ratios. In contrast, factors with the highest Sharpe ratios from our models deliver economically and statistically significant alphas relative to all traditional models. For IC-IPCA, three of the top four factors (ranked by Sharpe ratio) exhibit significant alphas across FF3, FF5, and Q4. For example, the first and second IC-IPCA factors earn monthly alphas of 0.17% and 0.15%, respectively, in the FF5 model, as shown in Panel A of Table 1.

The results are even stronger for DC-IPCA. The top two factors — Operating Illiquidity (OI) and Return Volatility (RV) — which play a central role in the tangency portfolios of our best-performing models (see Sections 5.5 and 5.6), generate substantial abnormal returns. For instance, OI earns a monthly alpha of 0.34% in the FF5 model, while RV earns 0.17%, both statistically significant at 1% as shown in Panel B of Table 1.

Taken together, these findings underscore that our clustering-based approach extracts economically meaningful and distinct sources of risk not captured by standard factor models ([Fama and French, 1993, 2015](#); [Hou et al., 2015](#)). By combining economic intuition with data-driven clustering, our method delivers interpretable factors that enhance pricing performance and provide new insights into the structure of expected returns.

5.3 Pricing of alpha and beta

Panel B of Table 1 reveals considerable heterogeneity in the Sharpe ratios of interpretable factors: while some exhibit high and statistically significant Sharpe ratios, others display limited explanatory power. This variation raises an important economic question about the underlying source of pricing ability. Specifically, does the predictive power of high-performing factors stem from compensation for systematic risk (*beta*), or does it reflect persistent mispricing captured

by non-zero pricing errors (*alpha*)? In other words, are these factors proxies for risk premia or anomalies outside the scope of standard models? Addressing this distinction is crucial for both theory and practice, as it informs whether the associated returns are justified by risk exposure or indicative of inefficiencies. To explore this, we proceed in two steps. First, we directly test for the presence of pricing errors (alphas) using asset-sorted portfolios. Second, we use a group-lasso–augmented C-IPCA framework to disentangle whether the pricing power of each cluster arises from alpha or beta.

Step 1: Testing for mispricing. We begin by estimating a C-IPCA model augmented with alphas at the end of each month t in the testing sample (January 2000–December 2021):

$$\mathbf{r}_t = \boldsymbol{\alpha}_t + \boldsymbol{\beta}_{t-1} \mathbf{f}_t + \mathbf{e}_t, \quad (23)$$

where $\boldsymbol{\alpha}_t \in \mathbb{R}^N$ captures time-varying pricing errors. We model alphas as a function of lagged firm characteristics:

$$\boldsymbol{\alpha}_t = \mathbf{X}_{t-1} \boldsymbol{\Gamma}_\alpha + \mathbf{v}_{t-1}, \quad (24)$$

where $\boldsymbol{\Gamma}_\alpha$ maps characteristics to alphas. All other notation in Equation (23) follows Equation (1). Using $\hat{\boldsymbol{\alpha}}_t$, we sort stocks into quintiles at the end of each month and form value-weighted portfolios. A long-short “alpha” portfolio takes a long position in the highest-alpha quintile and a short position in the lowest-alpha quintile. If alpha represents genuine mispricing, this long-short portfolio should earn a significant positive return.

Panel A of Table 2 confirms this intuition. Stocks in the lowest-alpha quintile earn 0.48% per month, while those in the highest-alpha quintile earn 0.95%. The resulting long-short spread is 0.48% per month, both statistically and economically significant, indicating that $\boldsymbol{\alpha}_t$ captures persistent mispricing.

Step 2: Identifying whether clusters drive alpha or beta. Next, we examine whether each cluster's pricing power arises from alpha or beta. We estimate Equation (23) with a group-lasso penalty on the alpha channel:

$$\min_{\Gamma, \Gamma_\alpha, \{\mathbf{f}_t\}_{t=1}^T} \sum_{t=1}^T (\mathbf{r}_t - \mathbf{X}_{t-1}\Gamma_\alpha - \mathbf{X}_{t-1}\Gamma\mathbf{f}_t)' \mathbf{W}_t (\mathbf{r}_t - \mathbf{X}_{t-1}\Gamma_\alpha - \mathbf{X}_{t-1}\Gamma\mathbf{f}_t) + \lambda \sum_{k=1}^K \|\Gamma_\alpha^{(k)}\|_2, \quad (25)$$

where the first term is the value-weighted mean squared error (notation follows Equations (3) and (23)) and the second term imposes an L2 group-lasso penalty on alpha loadings. Here, $\Gamma_\alpha^{(k)}$ denotes the alpha loadings for the k -th cluster, and $[\Gamma_\alpha^{(0)}, \Gamma_\alpha^{(1)}, \dots, \Gamma_\alpha^{(K)}] = \Gamma_\alpha$, with $\Gamma_\alpha^{(0)}$ corresponding to the intercept.

We employ group lasso to impose sparsity at the level of economically interpretable clusters rather than individual characteristics. This approach is motivated by the structure of the C-IPCA model: within each cluster, firm characteristics are selected to capture a common underlying economic mechanism. As such, testing whether an entire cluster contributes to pricing alpha offers a cleaner, more interpretable assessment of whether the associated economic channel reflects mispricing. The group-level penalty further enhances interpretability by allowing for selective inclusion or exclusion of whole economic themes, aligning with the model's objective of balancing parsimony with economic relevance.

To evaluate whether a cluster's pricing power originates from alpha, we first assess the statistical significance of the alpha loadings, $\Gamma_\alpha^{(k)}$. A significantly nonzero estimate indicates that the cluster contributes to mispricing. While informative about statistical relevance, this approach does not capture the economic importance of each cluster. To address this, we follow the variance decomposition method of Coleman (2017), which attributes the cross-sectional variation in $\hat{\alpha}_t$ to individual clusters in a non-negative, order-invariant, and additive manner.⁷

⁷Specifically, each month t , we randomly permute the order of clusters and sequentially add them into the regression $\hat{\alpha}_t = \gamma_0 + \gamma_1 X_{t-1}^{sub} + \varepsilon_t$, where X_{t-1}^{sub} includes characteristics from a subset of clusters. At each step, we record the incremental R^2 , reflecting the marginal contribution of the newly added cluster. We repeat this

Clusters that account for a large portion of the explained variance are interpreted as having economically meaningful alpha-driven pricing power.

To evaluate whether a cluster's pricing power arises from beta, we examine the Sharpe ratio of its associated model factor: high Sharpe ratios indicate beta-driven pricing power.

Panel B of Table 2 presents the results. Columns *Mean – MDD* report factor performance, Column *Cluster* identifies the characteristic cluster, Column *Explains Alpha* indicates whether the cluster's alpha loadings are nonzero, and Column *Explained Variance* reports the variance decomposition results.

There are several things worth mentioning. First, the factors with the highest Sharpe ratios — Operating Illiquidity (OI), the Market factor (Mkt), and Operating Efficiency (OE) — exhibit negligible alpha contributions. Their estimated alpha loadings are statistically insignificant, and their explained alpha variance is near zero. This suggests that their pricing power arises predominantly from compensation for systematic risk, not mispricing.

Second, clusters such as Momentum (Mom), Profitability (Prof), Growth (Gr), and Intangibles (Int) exhibit the opposite pattern: they deliver relatively low Sharpe ratios but contribute substantially to alpha. For instance, the Momentum cluster alone accounts for nearly 50% of the total alpha variance, with a significant alpha loading. Similarly, Profitability, Growth, and Intangibles each explain approximately 10% of the variation in $\hat{\alpha}_t$. These findings indicate that their pricing ability stems from persistent mispricing rather than from exposure to priced risk. Return Volatility (RV) lies somewhere in between. It achieves a moderate Sharpe ratio and a significant alpha loading, yet its contribution to alpha variance is modest (10%), suggesting a mix of mispricing and risk-based explanations.

Taken together, these results reveal a clean separation: certain interpretable factors capture systematic risk premia, while others reflect cross-sectional mispricing. This decomposition

procedure 1,000 times and average the marginal contributions across permutations and over time to obtain each cluster's average variance share.

highlights the dual role that firm characteristics can play in expected returns and underscores the value of our framework in disentangling alpha- versus beta-driven sources of return predictability.

5.4 Mean-variance efficiency

Thus far, we have improved the interpretability of the IPCA model by introducing a clustering structure. However, this added structure also imposes parameter restrictions, as specified in Equations (6) and (7). A natural concern is whether these restrictions might compromise the model's pricing performance. To address this issue, we evaluate the mean-variance efficiency of various factor models. If the clustering structure is well-specified—i.e., the restrictions are not unduly binding—then the pricing performance of the resulting model should be *at least as good as*, if not better than, that of benchmark models.

We evaluate the mean-variance efficiency of models by examining the out-of-sample performance of their corresponding tangency portfolios estimated using historical data. Specifically, we construct the tangency portfolio using model-implied factors, following a purely out-of-sample procedure: for each month t , we estimate the mean and covariance matrix of factor returns using data available up to time t , and then track the realized return of the tangency portfolio in month $t + 1$.

Formally, let $\mathbf{f}_t$ denote the vector of factor returns at time t , and let $\boldsymbol{\mu}_t$ and $\boldsymbol{\Sigma}_t$ represent the sample mean vector and covariance matrix of $\mathbf{f}_t$, estimated using data up to time t . The weights of the tangency portfolio are given by:

$$\mathbf{w}_t^* = c_t \boldsymbol{\Sigma}_t^{-1} \boldsymbol{\mu}_t,$$

where c_t is a scalar that normalizes the portfolio's volatility. Following [Kelly et al. \(2019\)](#), we

scale the weights c_t each period to make sure an *ex-ante* volatility of 1% per month. Specifically, c_t is chosen such that the historical volatility of the tangency portfolio's returns, $\{\mathbf{w}_t^* \mathbf{f}_\tau\}_{\tau=1}^{t-1}$, equals 1%. This scaling ensures comparability across models while preserving the tangency portfolio's Sharpe-optimal composition. The realized return on the tangency portfolio in month $t + 1$ is then:

$$R_{t+1}^{\text{TP}} = \mathbf{w}_t^* \mathbf{f}_{t+1}.$$

Table 1 reveals that not all factors extracted by the C-IPCA model command economically meaningful risk premia. As such, we implement a model selection procedure to retain only a subset of the most informative factors. We consider two approaches to factor selection.

5.4.1 Ordered model selection.

Our first approach to model specification selects factors based on their out-of-sample Sharpe ratios in the training period. Following [Stambaugh and Yuan \(2017\)](#), we treat the market factor as a baseline and sequentially augment it with the top $J - 1$ C-IPCA factors ranked by Sharpe performance. This procedure yields a family of J -factor models, which we refer to as the *Ordered C-IPCA* models. We provide the methodology of selecting factors in this subsection and the Section 5.5 shows the empirical results.

For the IC-based C-IPCA specification, we generate seven Ordered C-IPCA models corresponding to $J = 1, 2, \dots, 7$, reflecting the total number of interpretable factors identified in the IC clustering. Similarly, for the DC-based C-IPCA specification, we construct thirteen Ordered C-IPCA models for $J = 1, 2, \dots, 13$, based on the richer set of clusters derived from the data-driven clustering procedure.

Benchmark models. To evaluate the incremental value of the Ordered C-IPCA model, we compare its performance against the following benchmarks.

Benchmark 1: Standard IPCA. The first benchmark is the canonical IPCA model proposed by [Kelly et al. \(2019\)](#), estimated using the full set of firm characteristics without imposing any structural restrictions. For comparability, we retain the same number of factors J as in the corresponding Ordered C-IPCA to ensure that any observed differences are not simply due to the number of factors included.

Benchmark 2: Ordered IPCA. One might argue that comparing the Ordered C-IPCA with the first benchmark is not fair, as the Ordered C-IPCA model selects the J strongest factors (i.e., those with the highest Sharpe ratios) among all C-IPCA factors, whereas the canonical IPCA model does not perform any factor selection. To address this concern, we construct an additional benchmark that applies the same factor selection procedure as the Ordered C-IPCA model. Specifically, we first estimate a canonical IPCA model with the same total number of factors as the corresponding C-IPCA model (7 factors for the IC-IPCA model and 13 factors for the DC-IPCA model). We then select the J strongest factors from these 7 or 13 IPCA factors using the same selection method as in the Ordered C-IPCA model. The resulting models are referred to as Ordered IPCA7 and Ordered IPCA13, respectively. This ensures a fair comparison by holding the factor selection procedure constant between the Order CIPCA model and benchmarks.

5.4.2 Bayesian model selection

The ordered factor selection approach discussed in Section [5.4.1](#) focuses on the standalone performance of each factor, choosing those that perform best individually. However, factors often interact — some may be redundant, while others complement each other. To capture these joint effects and identify the most plausible combination of factors as a whole, we introduce a Bayesian model selection method developed by [Chib et al. \(2020\)](#).

Rather than comparing individual factors, the Bayesian model selection method provides a probabilistic way to compare competing models. Instead of testing models one by one, it

evaluates how likely each model is to be the “true” model given the observed data — this likelihood is expressed as a posterior probability. A model with a higher posterior probability is more consistent with the data and thus more likely to be the correct one.

In our context, this method can be used to select factors. Specifically, given a C-IPCA model with J factors ($J = 7$ for IC-IPCA and $J = 13$ for DC-IPCA). The J factors results in $L = 2^J - 1$ candidate models, each corresponding to a subset of the J factors. As shown in [Chib et al. \(2020\)](#), the Bayesian method can be used to estimate the posterior probability for each potential model, based on observed data. The 10 models with the highest posterior probabilities are then selected to form the Bayesian C-IPCA model.

Benchmark models. To evaluate the relative performance of the C-IPCA model, we compare it against the two following benchmark models.

Benchmark 1: Standard IPCA model. The first benchmark is the same as the first benchmark of the Ordered C-IPCA model, introduced in Section 5.4.1. It is the canonical IPCA model proposed by [Kelly et al. \(2019\)](#), estimated using the full set of firm characteristics without imposing any structural restrictions. To mitigate the influence of the number of factors, we use the canonical IPCA model with the same number of clusters as Bayesian IC-IPCA or Bayesian DC-IPCA models as benchmarks.

Benchmark 2: Bayesian IPCA Model. One might argue that comparing the Bayesian C-IPCA with the first benchmark is not entirely fair, since the Bayesian C-IPCA model represents the best model — i.e., the model with the highest posterior probability — among all candidate models formed from subsets of C-IPCA factors, whereas the canonical IPCA model does not perform any factor selection. To address this concern, we construct an additional benchmark that applies the same factor selection procedure as the Bayesian C-IPCA model. Specifically, we first estimate a canonical IPCA model with the same total number of factors as the corresponding C-IPCA model (7 factors for the IC-IPCA model and 13 factors for the DC-IPCA

model). We then identify the candidate model with the highest posterior probability among all possible subsets of 7 or 13 factors, using the same Bayesian selection procedure as in the Bayesian C-IPCA model. The resulting models are referred to as Bayesian IPCA7 and Bayesian IPCA13 models, respectively. This approach ensures a fair comparison by holding the model selection procedure constant between the Bayesian C-IPCA model and its benchmarks. We refer to these models as the Bayesian IPCA models.

5.5 Performance of Ordered C-IPCA Models

This subsection presents the empirical results on the performance of C-IPCA models using a subset of factors selected according to the ordered factor selection procedure. Specifically, we evaluate the out-of-sample Sharpe performance of tangency portfolios constructed from the top $J - 1$ C-IPCA factors — ranked by their Sharpe ratios in the training sample — together with the market factor.

Table 3 reports the annualized Sharpe ratios of these tangency portfolios for both the Ordered IC-IPCA (Panel A) and Ordered DC-IPCA (Panel B) specifications, across different values of J . For each model, we also provide results for two benchmark specifications discussed in Section 5.4.1: the standard IPCA model (Columns *IPCA* in both Panel A and Panel B) and the Ordered IPCA model (Columns *Ordered IPCA7* in Panel A and *Ordered IPCA13* in Panel B) with the same number of factors. Column *Clusters* lists the economic clusters associated with the newly selected C-IPCA factors, offering interpretability of the underlying factor structure.

There are several things worth mentioning via comparing the Panels A and B. *First*, Panel A demonstrates that the Ordered IC-IPCA model generally underperforms both the standard IPCA and Ordered IPCA models. Except for the smallest model sizes ($J = 1$ and $J = 2$), the OrderedIC-IPCA consistently exhibits lower Sharpe ratios. For instance, at $J = 3$, the Sharpe ratio of the OrderedIC-IPCA model is 0.67, compared to 0.86 for IPCA and 0.83 for OrderedIPCA

— gaps of 0.19 and 0.16, respectively. The performance deficit is most pronounced at $J = 6$, where the Ordered IC-IPCA's Sharpe ratio is 0.77 versus 1.19 for IPCA (gap of 0.42), and at $J = 5$, with a gap of 0.45 relative to Ordered IPCA. One likely explanation for this underperformance is the rigidity introduced by the intuitive clustering constraints: although economically motivated, such constraints may miss important cross-sectional correlations among firm characteristics that the data-driven clustering method is better suited to capture.

Second, Panel B shows that the Ordered DC-IPCA model consistently and significantly outperforms its Ordered IC-IPCA counterpart across all values of J . This performance dominance underscores the value of leveraging data-driven clustering to extract more accurate and informative groupings of characteristics. The DC clustering method captures latent correlation structures that are essential for identifying priced factors, thereby enhancing both model precision and empirical performance.

Third, focusing on the within-model performance of the Ordered DC-IPCA specification, we observe that the Sharpe ratio of the tangency portfolio improves steadily with J until peaking at $J = 7$, beyond which performance plateaus. This pattern suggests that the first seven factors contain the majority of pricing-relevant information, and adding more factors yields diminishing returns. This aligns with results in Table 1, which shows that only a subset of the factors have substantial risk premia, while others contribute little to cross-sectional pricing.

Finally, contrasting the Ordered DC-IPCA model with its benchmarks provides further insights. Across nearly all model sizes $J = 1, 2, \dots, 13$, the Ordered DC-IPCA outperforms the Ordered IPCA13 model, although the margin of improvement narrows as J increases. This suggests that the DC-IPCA framework not only improves interpretability but also delivers stronger (at least not worse) pricing performance than a similarly specified IPCA model. Similarly, the Ordered DC-IPCA model generally outperforms the standard IPCA benchmark, except for a

marginal underperformance at $J = 11$.⁸

Taken together, these findings highlight the empirical and practical value of combining economic structure with statistical discipline. The data-driven C-IPCA framework offers a compelling approach to constructing interpretable factor models that are also empirically efficient.

5.6 Performance of Bayesian C-IPCA models

This subsection presents the empirical results comparing model performance based on the Bayesian model selection procedure described in 5.4.2.

Figure 5 displays the posterior probabilities of the top 100 models with the highest posterior probabilities. Panel A of Figure 5 presents results for models based on the intuitive clusters. Within Panel A, the left subfigure shows the posterior probabilities of the top 100 models, while the right subfigure depicts the associated models. For clarity, we only display the top 5 most likely models in the right subfigure. Among the $2^7 - 1$ possible models, the most likely 5 models include (1) only one factor Momentum, (2) Momentum and Market, (3) Momentum, Market, Investment, and Intangible, (4) Momentum, Market, Investment, Intangible, and Value, and (5) Momentum, Market, and Investment.

Panel B of Figure 5 shows the results for models based on DC clusters. The left subfigure shows the posterior probabilities of the top 100 models, while the right subfigure highlights the top 5 models. Models include (1) Market, Momentum, Return Volatility, Operating Efficiency, Operating Illiquidity, and Size & Illiquidity, (2) Market, Momentum, Return Volatility, Operat-

⁸One important point is that Kelly et al. (2019) examines IPCA models with the number of factors $J = 1, 2, \dots, 6$, and find that the model with $J = 5$ achieves the highest Sharpe ratio. In contrast, in Panel B of our Table 4, the IPCA model with $J = 5$ rank only 10 in terms of Sharpe ratio. The main reason for this discrepancy lies in the weighting scheme used during estimation. As shown in Equation (3), we employ value weights in estimating the IPCA model, whereas Kelly et al. (2019) uses equal weights. Table B3 in Appendix B reports the results based on equal weighting using our dataset. In the testing sample, the Sharpe ratio of the $J = 5$ IPCA model under equal weights is 1.67, smaller only than the models with $J = 8, 9, 10$, which yield Sharpe ratios of 1.78, 1.86, and 1.93, respectively. Among the models with $J = 1, 2, \dots, 6$, the $J = 5$ model achieves the highest Sharpe ratio, consistent with the findings of Kelly et al. (2019).

ing Efficiency, and Operating Illiquidity, (3) Market, Momentum, Return Volatility, Operating Efficiency, Operating Illiquidity, Size & Illiquidity, and Profitability, (4) Market, Momentum, Return Volatility, Operating Efficiency, Operating Illiquidity, Investment, and (5) Market, Momentum, Return Volatility, Operating Efficiency, Operating Illiquidity, Value.

It is important to emphasize that the left subfigures in both Panel A and Panel B show that the most likely models exhibit much higher posterior probabilities than the remaining models. For instance, in the Bayesian IC-IPCA models, the top model delivers a posterior probability of 18%, followed by 12%, 6.5%, 5.8%, and 4.2% for the top 5 models. Similarly, in the Bayesian DC-IPCA models, the top two models deliver posterior probabilities of approximately 7.5% and 7.4%, which is significantly high considering that there are over 8,000 models. The following models have probabilities of 3.7%, 2.4%, and 2.3%. Overall, the posterior distribution indicates that the posterior probability is highly concentrated among the top models, thereby lending confidence to the selection of these models.

A natural and interesting question is the performance of the most likely models. Table 4 shows the performance, measured as the annualized Sharpe ratio of the tangency portfolio for each model. Specifically, Panel A and Panel B in Table 4 presents the performance of the top 10 models based on the DC and IC clusters, respectively. As a comparison, in each Panel, we also show results of the two benchmarks discussed in Section 5.4.2: standard IPCA models with the same number of factors and Bayesian IPCA models, respectively. Estimation of both benchmark models follows Section 5.4.2.

There are several key points to note. First, when comparing the performance of the tangency portfolios between Bayesian IC-IPCA and Bayesian DC-IPCA, we observe that the Bayesian DC-IPCA models consistently deliver higher Sharpe ratios. Specifically, excluding the sixth model, the Sharpe ratios of the Bayesian DC-IPCA models are all above 1.4, whereas the Sharpe ratios of the Bayesian IC-IPCA models remain below 0.80. This evidence suggests that incorpo-

rating data-driven information significantly boosts performance without compromising model interpretability.

When comparing the most likely models derived from the IPCA, we find that Bayesian DC-IPCA continues to deliver much higher Sharpe ratios than the IPCA models, which do not impose any restrictions on the characteristics coefficients of factor exposures.

Finally, when comparing the Bayesian DC-IPCA with the standard IPCA model with the same number of factor, we still find that our Bayesian DC-IPCA model delivers comparable or superior out-of-sample Sharpe ratios. In all cases, our model outperforms the standard IPCA model.

6 Mechanism and Robustness

Sections 5.5 and 5.6 demonstrate that the DC-IPCA models perform at least as well as and even better than the IPCA models no matter whether we use the ordered model selection or the Bayesian model selection to select a subset of factors (see details in Section 5.4.1 and 5.4.2). This section delves into the underlying mechanisms driving this model performance.

There are several factors which could boost the out-performance of our DC-IPCA models: information implied by the data, domain knowledge, and the parameter restriction. We discuss each of them in detail.

6.1 Value of data

Our DC clustering method combines both the domain knowledge (using the intuitive cluster as the starting point to make an initial partition as discussed in Section 2.2) and the characteristics similarity information implied by the data. The comparison between the DC-IPCA and IC-IPCA models indicates that our DC-IPCA model consistently and significantly out-performs

that of IC-IPCA in the out-of-sample Sharpe ratio, which indicates that beyond the domain knowledge, the information implied by the data provides valuable insights for the clustering.

6.2 Value of domain knowledge

Another potential explanation for the out-performance of our DC-IPCA method is the domain knowledge implied by the initial intuitive clusters. To evaluate the role of economic intuition, we construct alternative clusters using a purely data-driven approach - referred to as Purely Data-Driven Clusters (PDC). We compare the performance of Purely-Data-Driven C-IPCA, which is based solely on data-derived clusters, to DC-IPCA, which combines both economic intuition and data. This comparison allows us to assess the marginal value of incorporating economic information into the clustering process.⁹

To compare the model performance between Purely-Data-Driven C-IPCA and DC-IPCA, we follow the same procedure used to evaluate the DC-IPCA in Sections 5.5 and 5.6. Specifically, for each model, we first select a subset of factors according to the procedures described in Sections 5.4.1 and 5.4.2. We then construct the model's mean-variance efficient portfolio using the selected subset of factors and compare their performance.

Panel A of Table 5 presents the results. The first two columns report the performance of tangency portfolios based on the ordered factor selection approach outlined in Section 5.4.1. Specifically, factors are first ranked by their Sharpe ratio in the training sample, and then grad-

⁹Specifically, the Purely-Data-Driven Clusters are constructed by removing the economic information incorporated in the DC method, relying solely on data-derived clusters. In the DC method, economic information is incorporated by requiring that, within each sub-cluster, formed by splitting the set of firm characteristics, the characteristics must share the same economic explanation, or equivalently, belong to the same IC cluster. These economically coherent sub-clusters are then merged to form the final DC clusters in Section 3.3.2. In contrast, Purely-Data-Driven clustering removes the requirement that characteristics in each sub-cluster is within the same IC cluster. We obtain sub-clusters by partitions the whole set of firm characteristics. This comparison allows us to assess the marginal value of incorporating economic information into the clustering process. The optimal hyper-parameters for Purely Data-Driven Clustering are $m = 19, knn = 85, K = 10$. This means that Purely Data-Driven Clustering has 10 clusters and the corresponding Purely Data-Driven C-IPCA model is constructed following Section 2.2, with 11 factors (10 factors corresponding to 10 clusters and 1 ZC or market factor).

ually included based on their Sharpe ratio to construct J -factor ordered Purely Data-Driven C-IPCA models. The first column indicates the number of factors included, while the second column shows the Sharpe ratio of the tangency portfolios associated with the corresponding J -factor model.

The last two columns (the right panel *Bayesian Purely Data-Driven C-IPCA*) in Panel A displays the Sharpe ratios of the tangency portfolios based on the Bayesian model selection procedure outlined in Section 5.4.2. This procedure selects the 11 models with the highest posterior probability, ranking them by likelihood. The final column reports the number of factors in each model.

There are several findings worth mentioning in Panel A. First, the Sharpe ratios of the ordered Purely Data-Driven C-IPCA models are generally lower than those of the ordered DC-IPCA models (see Table 3). Except for the cases where $J = 1$ or $J = 4$, the ordered DC-IPCA model consistently delivers higher Sharpe ratios. For instance, when $J = 2$, the Sharpe ratio of the ordered Purely Data-Driven C-IPCA model is 0.74, which is 0.11 lower than the 0.85 achieved by the ordered DC-IPCA model. As J increases beyond 4, the gap in Sharpe ratios between the ordered Purely Data-Driven C-IPCA and ordered DC-IPCA models becomes even more pronounced. The highest Sharpe ratio for the ordered Purely Data-Driven C-IPCA model is 1.09 (when $J = 11$), whereas the ordered DC-IPCA model attains Sharpe ratios exceeding 1.4 when $J \geq 4$.

Second, similar trends are observed when comparing the Sharpe ratios based on the Bayesian model selection procedure. The Sharpe ratios of the Bayesian Purely Data-Driven C-IPCA models are consistently lower than those of the Bayesian DC-IPCA models (see Panel A in Table 4). Specifically, among all Bayesian Purely Data-Driven C-IPCA models, the highest Sharpe ratio is 1.07 (Rank = 5). In contrast, the lowest Sharpe ratio among the Bayesian DC-IPCA models is 1.09 (Rank = 6), which is still higher. Moreover, the second lowest Sharpe ratio from

the Bayesian DC-IPCA models is 1.41 (Rank = 9), significantly higher than the highest Sharpe ratio from the Bayesian Purely Data-Driven C-IPCA models (1.07).

Taken together, these results indicate that purely data-driven-based models consistently underperform DC-based models, underscoring the significant role that economic information plays in enhancing the performance of the DC-IPCA model.

6.3 Coefficient restrictions

We have demonstrated that both data and economic intuition contribute to the out-performance of the DC-IPCA model. However, one might argue that this improved performance is not solely due to the clustering structure itself but rather to the reduced number of parameters estimated. Specifically, the DC-IPCA model is constructed by imposing parameter restrictions derived from DC (see Equations (6) and (7)), effectively setting some parameters to zero and reducing the number of parameters to be estimated.

To assess whether the out-performance of the DC-IPCA model comes from its clustering structure or merely the smaller number of parameters, we consider a placebo test. Specifically, we generate random clusterings with the same number of clusters as the DC-IPCA model and construct a corresponding factor model denoted as Random-C-IPCA. Random clustering is generated without any reference to data or economic intuition, thus carrying no informational content implied by the data and domain knowledge. Instead, it takes a clustering structure that allows non-zero coefficients on characteristics within the cluster. As a result, the Random-C-IPCA model benefits from a similarly reduced number of parameters but lacks any meaningful clustering structure implied by the data and domain knowledge. If the out-performance of the DC-IPCA model is indeed attributable to its clustering structure driven by the data and intuition rather than parameter sparsity alone, we would expect the Random-C-IPCA to deliver a much inferior performance to the DC-IPCA model.

Panel B of Table 5 reports the performance comparison between the Random-C-IPCA models and the DC-IPCA model. We evaluate their performance using the same approach as described in Section 6.2. Specifically, for each model, we apply either the ordered or Bayesian model selection procedure, as outlined in Sections 5.4.1 and 5.4.2, to identify a subset of factors. We then compare the Sharpe ratios of the resulting models. The results corresponding to the two factor selection methods are presented in the left panel *Ordered-Random-C-IPCA* and the right panel *Bayesian-Random-C-IPCA*, respectively. Column J reports the number of selected factors, and the column “Rank” shows the model’s posterior probability ranking.

To mitigate concerns that the inherent randomness of the Random-C-IPCA might influence our results, we bootstrap 100 random C-IPCA. We then report the mean and median Sharpe ratios (columns *Sharpe–mean* and *Sharpe–median*) and the mean and median number of selected factors (columns *J–mean* and *J–median*).

There are several things worth mentioning in Panel B. First, generally speaking, the Sharpe ratios of the ordered Random-C-IPCA models are lower than those of the ordered DC-IPCA models (see Table 3). Specifically, with the exception of $J = 1$ and $J = 4$, the ordered DC-IPCA model consistently outperforms the ordered Random-C-IPCA model in terms of Sharpe ratios. For instance, at $J = 2$, the ordered Random-C-IPCA model yields an average and median Sharpe ratio of 0.66, which is 0.19 points below the 0.85 achieved by the ordered DC-IPCA model. As J exceeds 4, the performance disparity between the ordered Random-C-IPCA and ordered DC-IPCA models becomes even more pronounced. Although the ordered Random-C-IPCA model reaches its highest average and median Sharpe ratios of 1.19 (at $J = 12$) and 1.21 (at $J = 13$), respectively, the ordered DC-IPCA model surpasses 1.4 in both metrics once the number of factors exceeds 4.

Second, similar patterns emerge when comparing Sharpe ratios based on subsets of factors selected via the Bayesian procedure. The Sharpe ratios of the Bayesian Random-C-IPCA mod-

els are consistently lower than those of the Bayesian DC-IPCA models (see Table 4). Regardless of the rank of the Bayesian Random-C-IPCA model, its average (or median) Sharpe ratio never exceeds 0.6. In contrast, the lowest Sharpe ratio of the Bayesian DC-IPCA model is 1.09, and the second-lowest is 1.41 — both significantly higher than the average level achieved by the Bayesian Random-C-IPCA models.

Overall, models based on DC consistently outperform those based on random clustering, suggesting that the out-performance of the DC-IPCA model is due to its clustering structure rather than parameter sparsity alone.

6.4 Robustness test

6.4.1 Training sample

Our baseline analysis used the first 180 months as the training sample. To assess robustness, we extend the training window to 240, 300, 360, and 420 months. Table 6 reports the out-of-sample performance of Ordered DC-IPCA models alongside two benchmarks: (i) the standard IPCA model with the same number of factors and (ii) the Ordered IPCA models. For each training length, models are estimated in-sample, and tangency portfolio Sharpe ratios are computed out-of-sample. The first column lists the number of factors (J), and the remaining columns report annualized Sharpe ratios for the three model classes. The accompanying figure summarizes these results: the horizontal axis shows the number of factors, the vertical axis shows annualized Sharpe ratios, and bars with different colors correspond to different models.

Table 6 indicates that Ordered DC-IPCA generally outperforms both benchmarks. With a 240-month training sample, Ordered DC-IPCA consistently delivers higher Sharpe ratios than Ordered IPCA across all factor dimensions. Its performance relative to IPCA depends on model size: for larger J , Ordered DC-IPCA dominates, whereas for smaller J the difference is minimal. For example, with $J = 2$, Ordered DC-IPCA achieves a Sharpe ratio of 0.65 versus 0.70 for

IPCA. Results for the 300-month sample are similar. As the training window expands to 360 or 420 months, the advantage of Ordered DC-IPCA becomes more pronounced: across all factor counts, Ordered DC-IPCA achieves higher Sharpe ratios than both Ordered IPCA and IPCA.

Table 7 report the performance of the top 10 Bayesian DC-IPCA models with the highest posterior probabilities and compare them with the Bayesian IPCA benchmarks. Models are estimated using different training sample lengths, and we evaluate the out-of-sample tangency portfolio Sharpe ratios in the corresponding testing periods.

In each panel of Table 7, models are ranked by posterior probability. The first column reports the model rank, the second and third columns report the annualized Sharpe ratios in the testing sample, and number of factors included in each Bayesian C-IPCA model, respectively. The fourth and fifth columns show the annualized Sharpe ratios and number of factors for the Bayesian IPCA models.

Table 7 indicates that, for the same posterior probability rank, Bayesian DC-IPCA models achieve higher Sharpe ratios than Bayesian IPCA models while using fewer factors. This suggests that Bayesian DC-IPCA delivers superior predictive performance with a more parsimonious model structure.

Table 8 examines the performance of interpretable factors in DC-IPCA models across different training samples. The first column reports the factor IDs (MF), the second column lists annualized Sharpe ratios, and the third column shows the associated clusters. An asterisk (*) indicates that the factor is included in the top Bayesian DC-IPCA model as selected by Chib et al. (2024). Across all training samples, momentum-related factors — both fundamental momentum and return momentum — are consistently included in the top model with highest posterior probability. Factor selection varies with the training sample length: for the first 240 months, return volatility, growth of equity, and market beta are selected; for 300 months, R&D and illiquidity; for 360 months, operating illiquidity and return volatility; and for 420 months, market

beta, profitability, size, financial stability, growth, and value.

6.4.2 Model with equal weights

In the previous sections, we estimate the C-IPCA model by minimizing the *value-weighted* mean squared error, as defined in Equation (3), to mitigate the dominance of micro-cap stocks, as suggested by Hou et al. (2020). Under this specification, the estimated model factors correspond to the returns of value-weighted portfolios, as shown in Equation (4).

To assess the robustness of our results with respect to portfolio weighting schemes, we consider an alternative specification in which the model is estimated by minimizing the *equal-weighted* mean squared error. The empirical results for this specification are reported in Appendix B. The key message is that our main conclusions remain unchanged: combining economic intuition with data-driven clustering significantly enhances both model performance and interpretability. The objective function for the equal-weighted case is given by:

$$\min_{\mathbf{\Gamma}, \{\mathbf{f}_t\}_{t=1}^T} \sum_{t=1}^T (\mathbf{r}_t - \mathbf{X}_{t-1} \mathbf{\Gamma} \mathbf{f}_t)' (\mathbf{r}_t - \mathbf{X}_{t-1} \mathbf{\Gamma} \mathbf{f}_t), \quad (26)$$

where all notations follow those in Equation (3). The resulting model factors correspond to the returns of equal-weighted portfolios, as expressed in:

$$\hat{\mathbf{f}}_t = \left(\hat{\mathbf{\Gamma}}' \mathbf{X}_{t-1}' \mathbf{X}_{t-1} \hat{\mathbf{\Gamma}} \right)^{-1} \hat{\mathbf{\Gamma}}' \mathbf{X}_{t-1}' \mathbf{r}_t, \quad (27)$$

where notations are consistent with those in Equation (4).

To mitigate the disproportionate influence of low-priced stocks, we exclude observations with closing prices below \$5 when constructing equal-weighted portfolios, as such stocks tend to exhibit high volatility and limited liquidity, which can distort factor estimation. To maintain consistency between factor estimation and clustering, we also compute characteristic similarity

using equal-weighted correlations during the clustering stage:

$$\rho_{ij} = \frac{1}{T} \sum_{t=1}^T \frac{\sum_{n=1}^{N_t} (x_{it}^n - \bar{x}_{it})(x_{jt}^n - \bar{x}_{jt})}{\sqrt{\sum_{n=1}^{N_t} (x_{it}^n - \bar{x}_{it})^2} \sqrt{\sum_{n=1}^{N_t} (x_{jt}^n - \bar{x}_{jt})^2}}, \quad (28)$$

where $\bar{x}_{jt} = \frac{1}{N_t} \sum_{n=1}^{N_t} x_{jt}^n$, and all other notations follow those in Equation (17). All other methodological procedures remain identical to those described in the main text.

7 Conclusion

A vast literature documents numerous firm characteristics that explain the cross-section of stock returns. Treating each characteristic as a separate factor, however, leads to severe over-identification. Existing approaches — based on latent factor models or machine learning methods — reduce dimensionality by constructing factors as linear combinations of all characteristics. While these methods deliver a plausible number of factors, the resulting factors often lack clear economic interpretation.

This paper introduces a new framework that achieves both parsimony and interpretability in factor construction. Our approach proceeds in two steps. First, we cluster firm characteristics by combining economic intuition with data-driven similarity, producing statistically and economically coherent groups. Second, we extract a representative factor from each cluster that captures its underlying economic driving force. Applying this method to 94 characteristics from [Gu et al. \(2020\)](#), we identify a comparable number of factors like [Fama and French \(2015\)](#), each with a clear economic interpretation. Of the four most important factors, Operating Illiquidity (OI), Return Volatility (RV), Operating Efficiency (OE), and Size & Illiquidity (S&I), we find that one of them is related to famous models such as Fama-French 5-factor model ([Fama and French, 2015](#)). This helps to understand why the Fama-French 5-factor model and the alike fail

to explain the cross section of stock returns, as they do not capture fundamental arbitrage risks.

We further extend the influential IPCA model of Kelly et al. (2019) by incorporating a clustering structure, yielding the C-IPCA model. Clusters can be pre-specified based on economic theory or formed adaptively from the data. Empirically, C-IPCA retains the strong pricing performance of IPCA while delivering substantial gains in interpretability. Compared to the five latent factors in the original IPCA, our approach identifies economically transparent factors—such as illiquidity, momentum, investment, and market exposure—that perform as well or better. Moreover, by extending C-IPCA again to jointly estimate risk-based and mispricing components using a group-lasso penalty, we show that factors like OI, OE, and Mkt primarily capture systematic risk premia, while Momentum, Profitability, Growth, and Intangibles reflect persistent mispricing. These results highlight that embedding economic structure and clustering into IPCA not only enhances interpretability and predictive power but also reveals a clear separation between risk-driven and anomaly-driven factors.

Because factor models are fundamental to asset pricing, portfolio management, and risk analysis, our proposed framework has broad implications. For instance, Filippou et al. (2024) apply our approach to derive interpretable factors for currency returns. In addition, other factors, such as those identified by large language models, can also be readily integrated into our analysis. By bridging the gap between statistical efficiency and economic interpretability, the framework provides a practical and theoretically grounded tool for uncovering the fundamental sources of cross-sectional variation in expected returns.

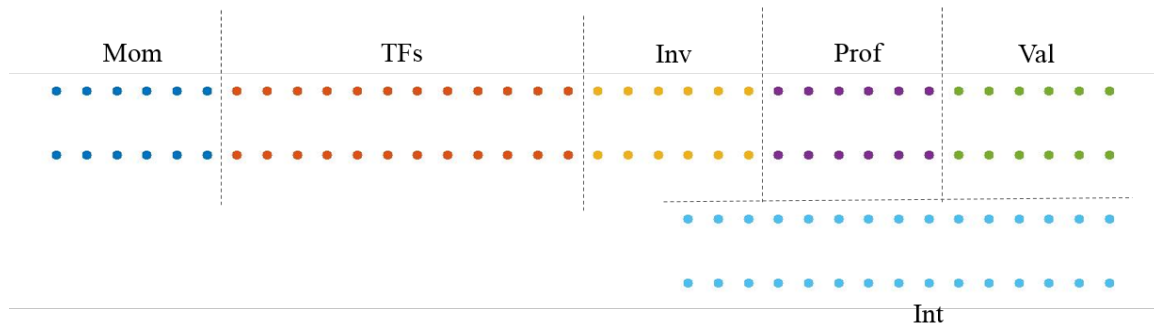
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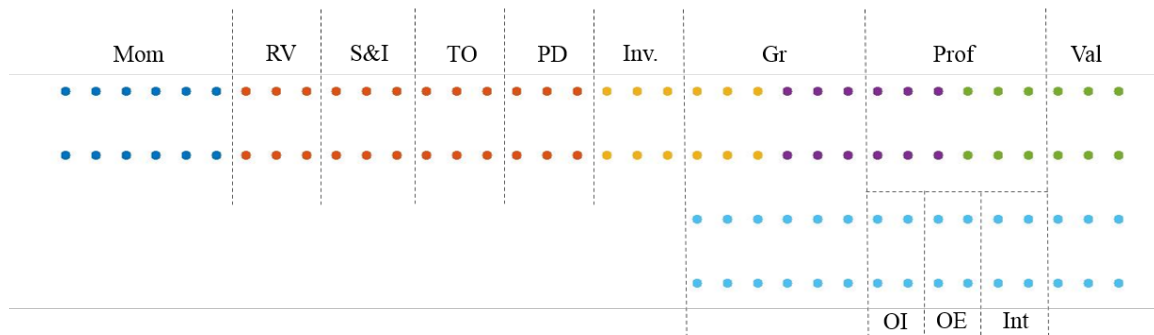
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8 Figures & Tables

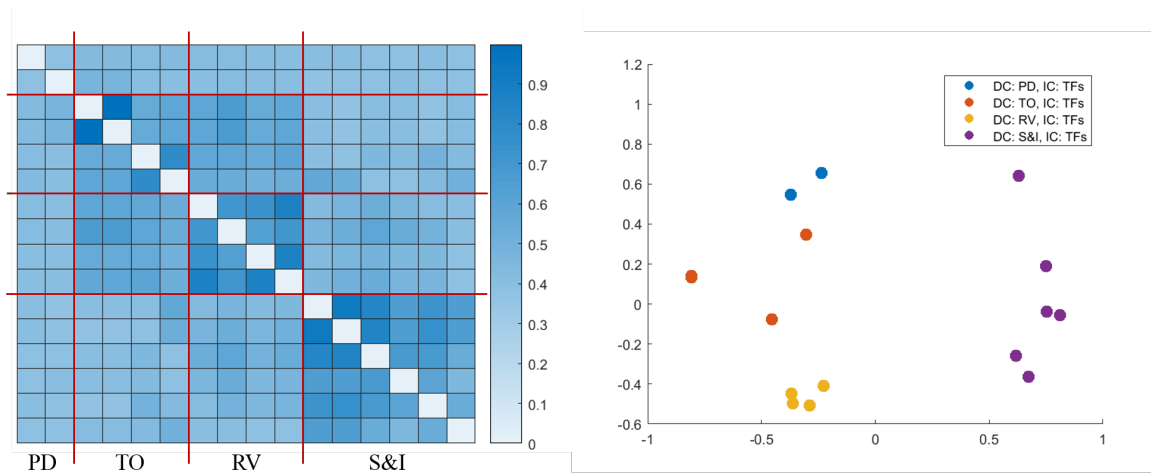


(a) IC

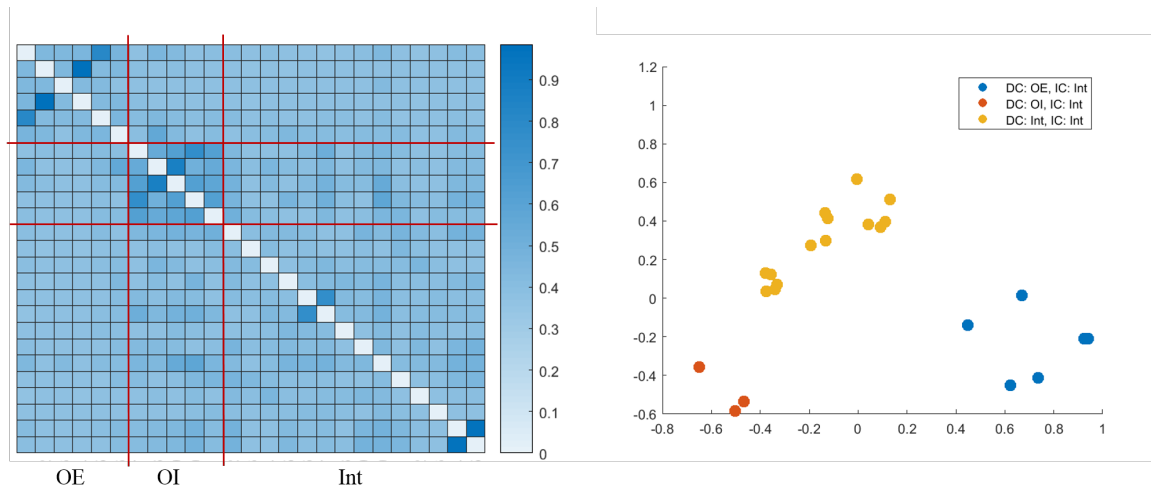


(b) DC

Figure 1: Cluster results. This figure illustrates the relationship between IC and DC clustering structures. Subfigure (a) shows the six IC clusters (Mom = Momentum, TFs = Trading Frictions, Inv = Investment, Prof = Profitability, Val = Value, Int = Intangibles), while Subfigure (b) displays the thirteen DC clusters (Mom = Momentum, RV = Return Volatility, S&I = Size & Illiquidity, TO = Turnover, PD = Price Delay, Inv = Investment, Gr = Growth, Prof = Profitability, Val = Value, OI = Operating Illiquidity, OE = Operating Efficiency, Int = Intangibles). Each vertex represents a firm characteristic, with colors indicating IC or DC cluster membership. The number of vertices is chosen for visualization purposes only and does not correspond to the actual count of firm characteristics. Dashed lines depict the clustering boundaries among characteristics.

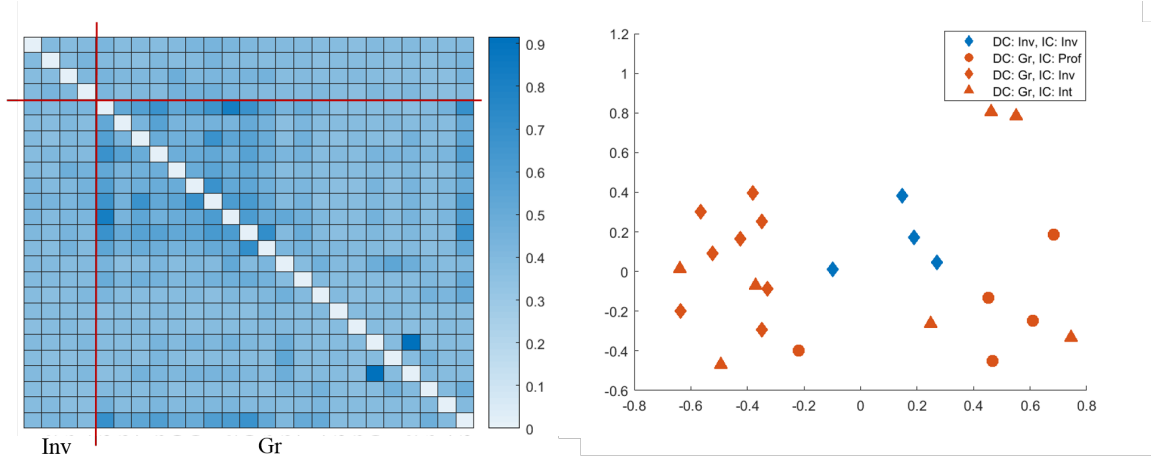


(a) RV, S&I, TO and PD clusters in DC

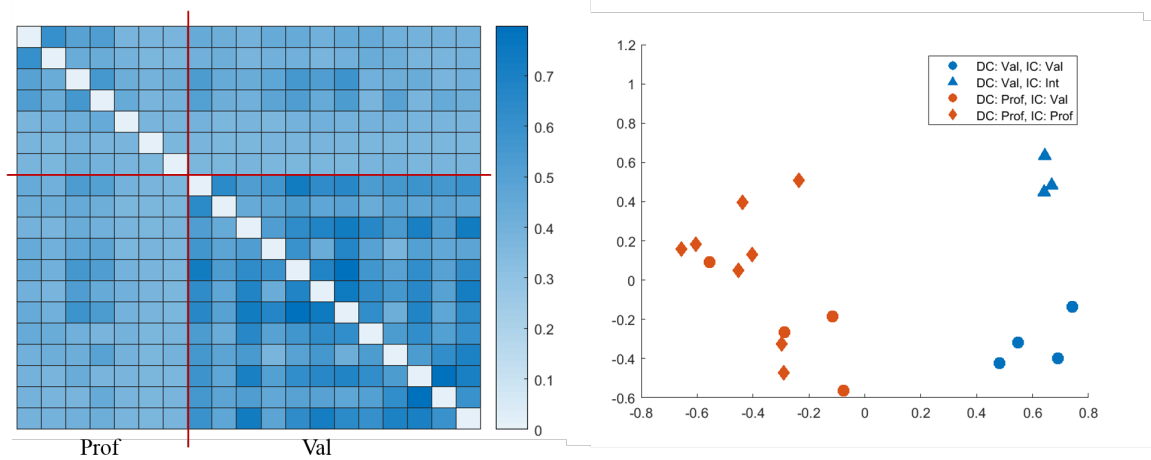


(b) OE, OI and Int clusters in DC

Figure 2: Role of data information in clustering. This figure illustrates the role of data information in clustering. Subfigure (a) focuses on four DC trading-friction clusters: RV, S&I, TO, and PD. Subfigure (b) highlights three DC intangibles-related clusters: OE, OI, and Int. (See Figure 1 for cluster abbreviations.) In each subfigure, the left panel shows the similarity matrix $s_{i,j}$, where rows and columns represent firm characteristics, and cell shading indicates pairwise similarity (darker cells denote higher similarity). Red lines delineate DC clusters. The right panel visualizes firm characteristics in two dimensional space using Multidimensional Scaling (MDS) algorithm following Borg (2011), based on distances defined as $d_{i,j} = 1/s_{i,j} - 1$. The MDS maps each firm characteristic into a point in a two dimensional space while keep the distance between characteristics roughly unchanged. Vertices represent firm characteristics, with colors denoting DC clusters and shapes indicating IC clusters, enabling direct comparison between clustering approaches..

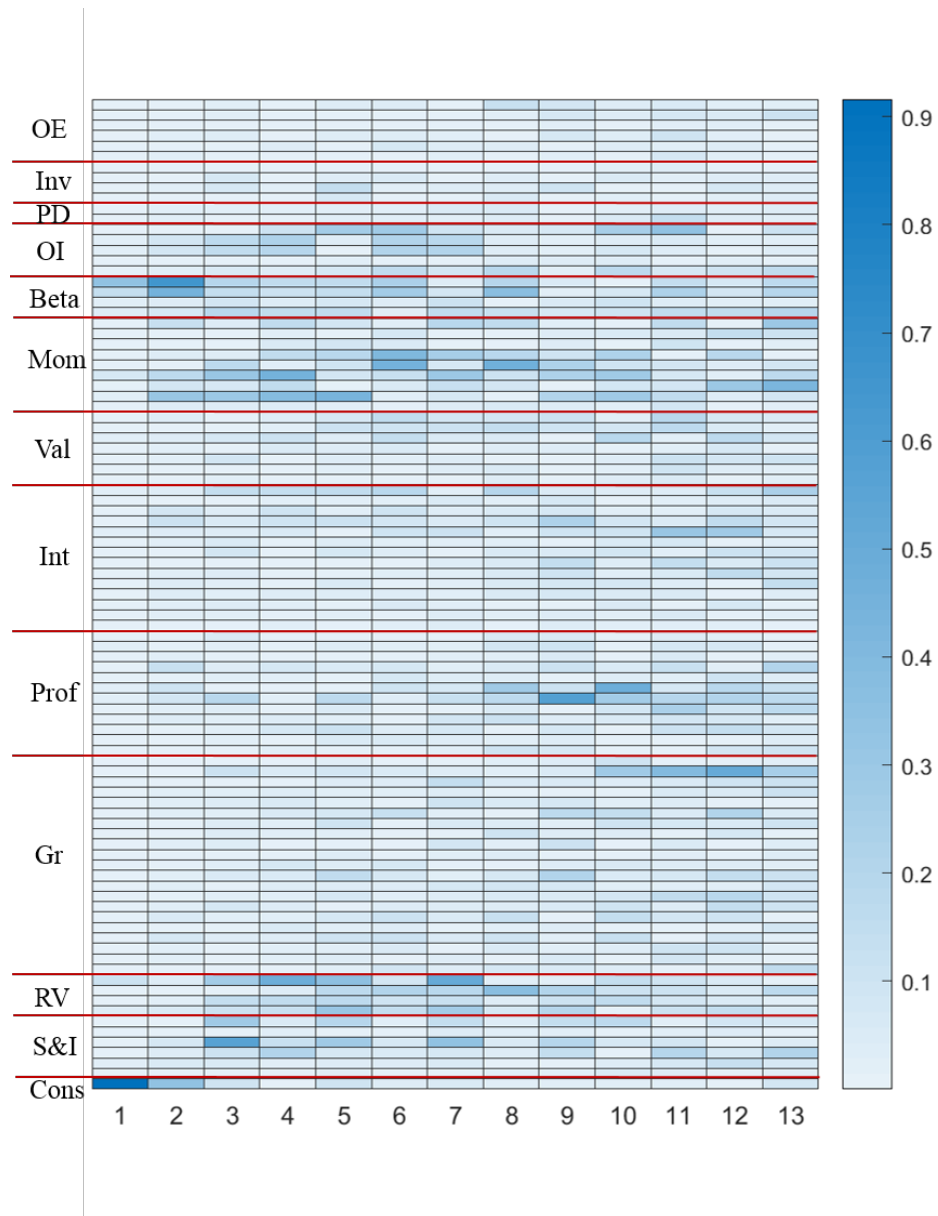


(a) Inv and Gr clusters in DC



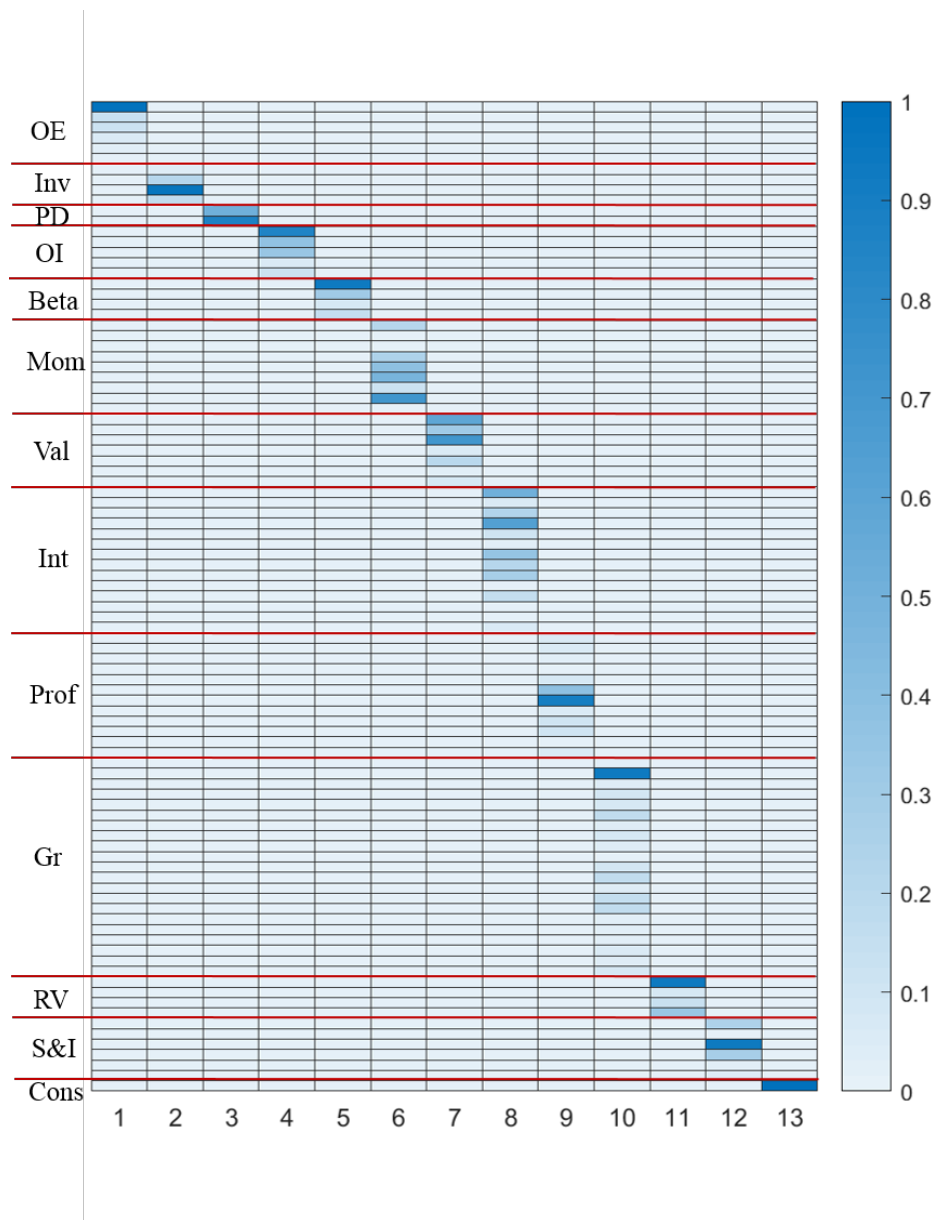
(b) Val and Prof clusters in DC

Figure 3: **Role of data information in clustering (cont).** This figure examines how data information influences clustering outcomes. Subfigure (a) focuses on the Inv and Growth clusters in DC, while Subfigure (b) highlights the Val and Prof clusters in DC. (See Figure 1 for cluster abbreviations.) In each panel, the left subfigure displays the similarity matrix $s_{i,j}$, where rows and columns correspond to firm characteristics, and cell shading reflects similarity (darker colors indicate higher similarity). The right subfigure visualizes the spatial distribution of characteristics using Multidimensional Scaling (MDS) following Borg (2011), based on distances defined as $d_{i,j} = 1/s_{i,j} - 1$. The MDS maps each firm characteristic into a point in a two dimensional space while keep the distance between characteristics roughly unchanged. Vertices represent firm characteristics, with colors denoting DC clusters and shapes indicating IC clusters, enabling direct comparison between clustering approaches.



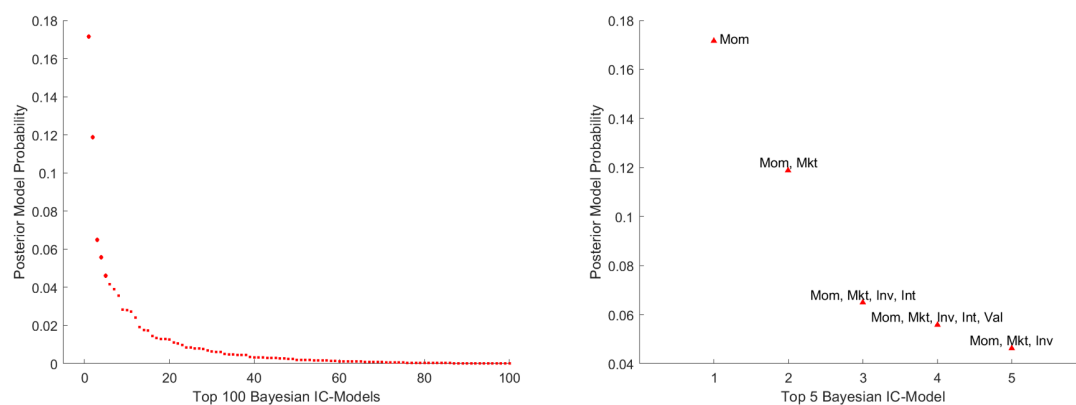
(a) IPCA

Figure 4: **The Gamma matrix $\hat{\Gamma}$ for models**

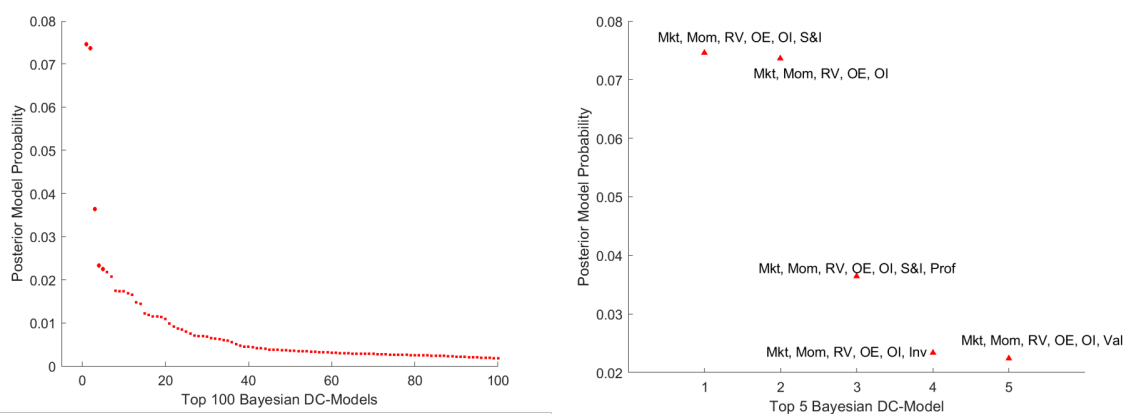


(b) C-IPCA

Figure 4: **The Gamma matrix $\hat{\Gamma}$ for models (cont.).** This figure presents the absolute values of the $\hat{\Gamma}$ matrix from equation (2) for the IPCA model with 13 factors (Panel (a)) and the DC-IPCA model (Panel (b)), estimated using the full sample. Each column corresponds to a factor in f_t (or an exposure in β_t) in equation (1), while each row represents a firm characteristic. Red lines partition the rows into 12 clusters based on DC and one constant. Each cell represents the absolute loading of characteristic i on exposure j , reflecting the characteristic's importance for that factor. Darker shading indicates higher absolute values and thus greater importance. To enhance comparability across factors, each column is scaled so that the sum of squared elements equals one.



(a) Bayesian IC-IPCA models



(b) Bayesian DC-IPCA models

Figure 5: Posterior probability of Bayesian C-IPCA models. This figure displays the posterior probabilities of Bayesian C-IPCA models. For all candidate models—each representing a subset of factors from a given C-IPCA specification—we compute and rank posterior probabilities. The ranked probabilities are shown. Subfigures (a) and (b) correspond to two versions of Bayesian C-IPCA models based on different clustering schemes: Bayesian IC-IPCA and Bayesian DC-IPCA, respectively. In each subfigure, the left panel reports the most likely 100 Bayesian C-IPCA models, where the x-axis indicates the posterior probability rank and the y-axis shows the posterior probability. The right panel highlights the most likely five Bayesian C-IPCA models along with their associated clusters. (See Figure 1 for cluster abbreviations; “Mkt” denotes the zero-correlation market factor (ZC).)

Table 1: Performance of interpretable factors

This table reports the performance of interpretable factors from the IC-IPCA model (Panel A) and the DC-IPCA model (Panel B). Each row corresponds to an individual factor. The first column lists factor IDs. To facilitate comparison, factors are ranked by their out-of-sample Sharpe ratios over the testing period (2000:01–2021:12), except for the market factor, which is reported in the final row. Columns two through five present performance metrics of each factor portfolio: sample mean (Mean), sample standard deviation (SD), annualized Sharpe ratio (Sharpe), and maximum drawdown (MDD) of monthly returns. Columns six through eight present the alpha of regressing model factors on traditional factor models, including Fama French 3 factor model (FF3, [Fama and French, 1993](#)), Fama French 5 factor model (FF5, [Fama and French, 2015](#)) and q-factor factor model (Q4, [Hou et al., 2015](#)). *, **, and *** indicate statistical significance of alpha at the 10%, 5%, and 1% confidence levels, respectively, based on Newey-West adjusted standard errors ([Newey and West, 1986](#)). The final column indicates the associated clusters (see [Figure 1](#) for abbreviations; “Mkt” refers to the zero-correlation market factor (ZC)). Panel C reports correlations between the market factor (MktRf) and the following: (i) the ZC factors from the IC-IPCA and DC-IPCA models, and (ii) the factor from the IPCA model — of the same number of factors as IC-IPCA and DC-IPCA — that exhibits the highest correlation with the market factor (denoted as MF(IPCA7) and MF(IPCA13), respectively). MktRf is computed as the excess return on the value-weighted portfolio. The sample period spans January 2000 to December 2021.

MFs	Mean (%)	SD (%)	Sharpe	MDD	alpha-FF3	alpha-FF5	alpha-Q4	Econ. Interp.
Panel A. IC-IPCA								
1	0.19	1.06	0.62	7.93	0.17***	0.17***	0.14**	Inv
2	0.11	0.87	0.44	10.02	0.13**	0.15**	0.17***	Int
3	0.09	0.87	0.35	8.77	0.06	-0.02	0.02	Val
4	0.08	1.11	0.25	10.81	0.18***	0.11**	0.15***	TFs
5	0.07	1.04	0.23	16.96	0.1	-0.01	0.09	Prof
6	0.07	1.15	0.20	18.70	0.11	0.05	0.15**	Mom
7	0.14	1.03	0.46	15.12	0.00	0.00	0.00	Mkt
Panel B. DC-IPCA								
1	0.24	1.14	0.73	8.59	0.25***	0.34***	0.29***	OI
2	0.21	1.27	0.56	8.80	0.30***	0.17***	0.24***	RV
3	0.10	0.98	0.37	11.67	0.09*	0.06	0.09	OE
4	0.08	0.93	0.30	23.06	-0.08	-0.05	-0.09	S&I
5	0.07	0.86	0.29	8.96	0.08	0.06	0.05	Int
6	0.08	0.95	0.29	9.80	0.07	0.08	0.05	Gr
7	0.06	1.12	0.20	17.86	0.11	0.05	0.15**	Mom
8	0.05	1.00	0.17	17.15	-0.10*	0.01	-0.12*	Prof
9	0.05	1.07	0.16	10.13	0.03	0.08	0.04	Turn
10	0.04	1.12	0.13	16.78	-0.04	-0.02	-0.03	PD
11	0.02	0.98	0.08	11.78	0.02	0.02	0.02	Inv
12	0.00	0.92	0.02	12.92	0.01	0.01	0.00	Val
13	0.14	1.03	0.46	15.12	0.00	0.00	0.00	Mkt
Panel C. corr. of market factors								
ZC(IC-IPCA)	1.00							
ZC(DC-IPCA)	1.00	1.00						
MF(IPCA7)	0.99	0.99	1.00					
MF(IPCA13)	0.99	0.99	1.00	1.00				
MktRf	1.00	1.00	0.99	0.99	1.00			

Table 2: Clusters related to alpha and beta

The table distinguishes clusters related to alpha and beta. Panel A reports the performance of alpha portfolios. The alpha portfolios are constructed as follows. First, at the end of each month t in the testing sample, we estimate the C-IPCA model with alpha, as specified in equation (23), using all available data through month t . Stocks are then sorted into quintiles based on the estimated alpha. Panel A reports the average returns and corresponding t-statistics for the five portfolios and the long-short portfolio. The t-statistics are based on Newey-West adjusted standard errors (Newey and West, 1986). Panel B separates clusters whose pricing power is driven by alpha from those driven by beta. Specifically, we estimate the C-IPCA model with alpha using the full sample, while augmenting the objective function with a penalty term, as in equation (25), so that the alpha loadings of certain clusters of firm characteristics are shrunk to zero. The first five columns of Panel B show the performance of model factors, f_i in equation (23), and the sixth column reports the corresponding clusters. The seventh column indicates whether alpha loadings of the cluster are nonzero. The eighth column reports results of the variance decomposition of alpha. In particular, we decompose the variation of alpha to measure how much of it is explained by each cluster of firm characteristics, following the method of Coleman (2017).

Panel A. Performance of alpha portfolios

	L	2	3	4	H	H-L
Mean (%)	0.48	0.92	1.16	0.976	0.956	0.486
t-stat	1.21	2.52	3.24	3.28	3.72	2.37

Panel B. Separating alpha and beta

MFs	Mean (%)	SD (%)	Sharpe	MDD	Cluster	Explains Alpha	Explained Variance (%)
1	0.39	1.85	0.74	17.19	OI	No	0.86
2	0.16	1.00	0.57	14.69	Mkt	No	0.00
3	0.90	6.62	0.47	63.51	OE	No	0.46
4	0.28	2.09	0.47	30.87	RV	Yes	10.06
5	0.72	10.07	0.25	83.73	Inv	No	0.39
6	0.14	2.07	0.24	37.57	S&I	No	1.79
7	0.25	4.07	0.21	58.46	Mom	Yes	49.53
8	0.26	6.80	0.13	84.25	Val	Yes	1.76
9	0.09	3.61	0.09	63.22	Prof	Yes	12.53
10	0.09	3.56	0.08	51.79	Gr	Yes	10.90
11	0.46	22.91	0.07	100.00	PD	No	0.19
12	0.01	1.28	0.04	25.74	Turn	No	1.40
13	0.03	2.85	0.03	56.00	Int	Yes	10.12

Table 3: Mean-variance efficiency of the Ordered C-IPCA, IPCA and Ordered IPCA models

This table reports the out-of-sample Sharpe ratios of tangency portfolios for Ordered C-IPCA models and two benchmarks: (i) the IPCA model with the same number of factors, and (ii) the Ordered IPCA models. The J factor Ordered C-IPCA model consists of the first $J - 1$ factors from the corresponding C-IPCA specification and the market factor, with factors ordered by their Sharpe ratios in the training sample (1985:01–1999:12). The Ordered IPCA benchmark for a given Ordered C-IPCA model includes the first J factors from an IPCA model. Specifically, the Ordered IPCA model paired with Ordered IC-IPCA (Panel A) is based on an IPCA model with 7 factors (Ordered IPCA7), while the benchmark for Ordered DC-IPCA (Panel B) is based on an IPCA model with 13 factors (Ordered IPCA13). Tangency portfolios are constructed entirely out-of-sample by estimating the mean and covariance matrix of model factors using data up to time t and computing the portfolio return at $t + 1$. The first column lists the number of factors (J). Columns two through four report the annualized Sharpe ratios for Ordered C-IPCA, IPCA, and Ordered IPCA models, respectively. The final column identifies the cluster associated with the newly added factor in the Ordered C-IPCA model. (See Figure 1 for cluster abbreviations; “Mkt” refers to the zero-correlation market factor (ZC).) The sample period spans January 2000 through December 2021.

Panel A. IC				
J	Ordered IC-IPCA	IPCA	Ordered IPCA7	Clusters
1	0.43	0.31	0.37	Mkt
2	0.50	0.59	0.44	Mom
3	0.67	0.86	0.83	Inv
4	0.77	0.88	0.90	Int
5	0.75	0.86	1.20	Prof
6	0.77	1.19	1.19	Val
7	0.88	1.23	1.23	TFs
Panel B. DC				
J	Ordered DC-IPCA	IPCA	Ordered IPCA13	Clusters
1	0.43	0.31	0.21	Mkt
2	0.85	0.59	0.34	OI
3	0.84	0.86	0.45	OE
4	0.83	0.88	0.77	Mom
5	1.46	0.86	0.73	RV
6	1.44	1.19	1.00	Inv
7	1.49	1.23	1.31	Gr
8	1.49	1.24	1.32	PD
9	1.46	1.20	1.29	Prof
10	1.47	1.28	1.27	S&I
11	1.46	1.51	1.26	Int
12	1.47	1.41	1.38	TO
13	1.45	1.35	1.35	Val

Table 4: Mean-variance efficiency of the Bayesian C-IPCA, IPCA and Bayesian IPCA models

This table reports the out-of-sample Sharpe ratios of tangency portfolios for the top 10 Bayesian DC-IPCA models (Panel A) and Bayesian IC-IPCA models (Panel B) with the highest posterior probabilities. In each Panel, we also show the two benchmarks for the Bayesian C-IPCA models: Bayesian IPCA models and standard IPCA models with the same number of factors, estimated following Section 5.4.2. For each model, we show Sharpe ratio of their tangency portfolios (Sharpe) and the number of factors (# factors). All tangency portfolios are constructed on a purely out-of-sample basis, using all observations up to time t to estimate the mean and covariance matrix, with portfolio returns evaluated at $t + 1$. The sample period spans January 2000 through December 2021.

		Top1	2	3	4	5	6	7	8	9	10
Panel A. DC-IPCA											
Bayesian DC-IPCA	Sharpe	1.44	1.46	1.42	1.44	1.42	1.09	1.45	1.45	1.41	1.43
	# factors	6.00	5.00	7.00	6.00	6.00	4.00	4.00	6.00	7.00	7.00
Bayesian IPCA13	Sharpe	0.54	0.82	0.96	0.85	0.52	0.34	1.10	0.80	0.64	1.06
	# factors	4.00	5.00	6.00	6.00	5.00	3.00	7.00	6.00	5.00	8.00
IPCA	Sharpe	1.19	0.86	1.23	1.19	1.19	0.88	0.88	1.19	1.23	1.23
	# factors	6.00	5.00	7.00	6.00	6.00	4.00	4.00	6.00	7.00	7.00
Panel B. IC-IPCA											
Bayesian IC-IPCA	Sharpe	0.21	0.50	0.77	0.79	0.67	0.39	0.63	0.48	0.33	0.62
	# factors	1.00	2.00	4.00	5.00	3.00	2.00	3.00	2.00	2.00	3.00
Bayesian IPCA7	Sharpe	0.83	0.44	0.37	0.73	0.74	0.90	0.32	0.88	1.16	0.82
	# factors	3.00	2.00	1.00	2.00	2.00	4.00	1.00	4.00	4.00	4.00
IPCA	Sharpe	0.31	0.59	0.88	0.86	0.86	0.59	0.86	0.59	0.59	0.86
	# factors	1.00	2.00	4.00	5.00	3.00	2.00	3.00	2.00	2.00	3.00

Table 5: Mean-variance efficiency of the Purely-Data-Driven C-IPCA and Random-C-IPCA models

This table reports the mean-variance efficiency of the Purely-Data-Driven C-IPCA and Random-C-IPCA models using ordered or bayesian model selection. Purely-Data-Driven clustering refers to clusters derived exclusively from data, without incorporating economic information from the IC prior. Random clustering denotes clusters generated entirely at random, without using any economic or data-driven inputs. Panel A presents results for the Ordered (Bayesian) -Purely-Data-Driven-C-IPCA models, while Panel B reports results for the Ordered (Bayesian)-Random-C-IPCA models. Within each panel, the left set of columns shows results based on ordered model selection, and the right set shows results based on Bayesian model selection. Column *J* denotes the number of model factors; *Sharpe* reports the tangency portfolio Sharpe ratio; and *Rank* gives the ranking of bayesian model selection. For the Ordered(Bayesian)-Random-C-IPCA models, *Mean* and *Median* report the average and median results across 100 random clustering iterations.

Panel A. Performance of Purely-Data-Drive-C-IPCA							
Ordered-Purely-Data-Driven C-IPCA			Bayesian-Purely-Data-Driven C-IPCA				
J	Sharpe		Rank	Sharpe		J	
1	0.43		1	0.87		5	
2	0.74		2	0.69		4	
3	0.69		3	0.76		3	
4	0.94		4	0.83		6	
5	0.87		5	1.07		6	
6	0.83		6	0.94		4	
7	0.86		7	0.72		4	
8	0.88		8	0.65		5	
9	0.85		9	0.89		6	
10	0.90		10	0.84		6	
11	1.09		11	0.89		6	

Panel B. Average performance of Random-C-IPCA							
Ordered-Random-C-IPCA			Bayesian-Random-C-IPCA				
J	Sharpe		Rank	Sharpe		J	
	Mean	Median		Mean	Median	Mean	Median
1	0.43	0.43	1	0.54	0.54	4.83	4
2	0.66	0.66	2	0.49	0.50	4.61	4.5
3	0.80	0.78	3	0.52	0.53	4.71	5
4	0.90	0.89	4	0.52	0.52	4.71	5
5	0.98	0.96	5	0.53	0.53	5.01	5
6	1.02	1.00	6	0.57	0.57	5.16	5
7	1.05	1.02	7	0.56	0.55	5.15	5
8	1.08	1.07	8	0.55	0.56	5.13	5
9	1.10	1.11	9	0.53	0.55	5.24	5
10	1.12	1.11	10	0.56	0.56	5.13	5
11	1.13	1.12	11	0.52	0.53	5.12	5
12	1.16	1.16	12	0.55	0.52	5.28	5
13	1.19	1.21	13	0.56	0.56	5.20	5

Table 6: Different training samples: Mean-variance efficiency of the Ordered DC-IPCA, IPCA and Ordered IPCA models

This table reports the out-of-sample Sharpe ratios of tangency portfolios for Ordered DC-IPCA models and two benchmarks: (i) the IPCA model with the same number of factors, and (ii) the Ordered IPCA models. The length of the training sample varies from 240 to 420 months. The first column lists the number of factors (J), while the remaining columns report annualized Sharpe ratios for Ordered DC-IPCA, IPCA, and Ordered IPCA models in the testing sample (the full sample excluding the training period). The bottom figure illustrates these results: the horizontal axis represents the number of factors, and the vertical axis represents annualized Sharpe ratios. Bars with different colors indicate different models.

	Train: 240 months			Train: 300 months			Train: 360 months			Train: 420 months		
	Ordered DC-IPCA	IPCA	Ordered IPCA	Ordered DC-IPCA	IPCA	Ordered IPCA	Ordered DC-IPCA	IPCA	Ordered IPCA	Ordered DC-IPCA	IPCA	Ordered IPCA
1	0.59	0.53	0.17	0.92	0.81	0.23	0.84	0.78	0.36	1.01	1.05	-0.08
2	0.65	0.70	0.26	0.95	1.09	0.15	1.14	0.87	0.17	0.83	0.69	-0.75
3	0.83	0.95	0.29	0.85	1.34	0.55	1.47	1.14	0.46	1.57	1.09	-0.82
4	0.84	1.07	0.56	1.27	1.53	0.65	1.79	1.42	0.65	2.09	1.46	-0.57
5	0.98	1.13	0.88	1.44	1.35	0.51	2.21	1.38	1.10	2.04	1.73	-0.06
6	0.97	1.22	0.77	1.52	1.60	0.55	2.26	1.30	1.82	2.41	1.05	0.69
7	1.44	1.13	0.91	1.47	1.61	0.61	2.27	1.14	2.02	1.85	0.88	0.91
8	1.42	0.97	0.87	1.99	1.45	1.10	2.21	1.14	1.92	1.70	0.73	0.76
9	1.41	1.12	1.01	1.96	1.54	1.32	2.21	1.42	1.86	1.78	1.74	0.62
10	1.33	1.11	1.01	1.94	1.81	1.28	2.17	1.81	1.83	1.42	1.51	0.76

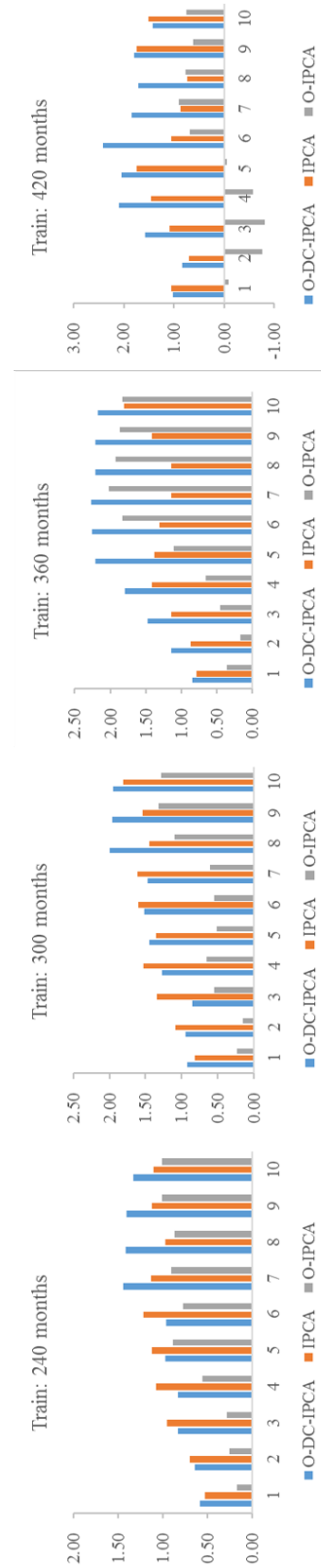


Table 7: Different training samples: Mean-variance efficiency of the Bayesian C-IPCA and Bayesian IPCA models

This table reports the out-of-sample Sharpe ratios of tangency portfolios for the top 10 Bayesian C-IPCA models with the highest posterior probabilities, estimated using different training sample lengths. Panels A through D correspond to training periods of 240, 300, 360, and 420 months, respectively. Within each panel, the first column indicates the model rank, the second and third columns report the annualized Sharpe ratios in the testing sample, and number of factors included in each Bayesian C-IPCA model. The fourth and fifth columns show the annualized Sharpe ratios and number of factors for the Bayesian IPCA models.

Panel A. Training Sample: 240 months					Panel B. Training Sample: 300 months				
Rank	Bayesian C-IPCA		Bayesian IPCA		Rank	Bayesian C-IPCA		Bayesian IPCA	
	Sharpe	#Factors	Sharpe	#Factors		Sharpe	#Factors	Sharpe	#Factors
1	1.13	6	1.05	8	1	1.80	5	1.23	8
2	1.37	7	1.01	9	2	1.96	6	1.12	8
3	1.13	7	1.03	9	3	1.89	6	1.17	7
4	1.20	5	0.91	7	4	2.02	7	1.28	8
5	1.12	7	1.08	9	5	1.74	6	1.32	9
6	1.16	7	1.06	9	6	1.76	6	1.24	9
7	1.12	7	1.04	9	7	1.82	6	1.18	9
8	1.13	7	0.99	10	8	1.80	6	0.77	7
9	1.10	7	1.01	10	9	1.98	7	1.53	10
10	1.45	6	0.87	8	10	1.92	7	1.28	10

Panel C. Training Sample: 360 months					Panel D. Training Sample: 420 months				
Rank	Bayesian C-IPCA		Bayesian IPCA		Rank	Bayesian C-IPCA		Bayesian IPCA	
	Sharpe	#Factors	Sharpe	#Factors		Sharpe	#Factors	Sharpe	#Factors
1	2.11	5	1.31	10	1	1.03	8	0.95	9
2	2.16	6	1.30	9	2	1.34	7	0.76	10
3	2.13	6	1.32	11	3	1.85	7	1.10	8
4	2.16	7	1.30	10	4	1.38	9	0.76	8
5	2.09	6	1.15	9	5	1.50	8	0.91	7
6	2.14	6	1.16	9	6	1.20	7	0.62	9
7	2.09	6	1.13	8	7	1.70	8	0.93	9
8	2.10	6	1.27	9	8	1.35	9	0.47	9
9	1.90	4	1.33	9	9	1.52	6	0.63	8
10	1.87	4	1.16	8	10	1.67	8	0.78	8

Table 8: Different Training Samples: Performance of DC-IPCA Model Factors

This table shows the performance of interpretable factors from the DC-IPCA models across different training sample lengths. The first column lists the IDs of model factors (MFs). To facilitate interpretation, factors are ranked by their Sharpe ratios in the testing sample, except for the market factor, which is reported in the final row. The second column presents the annualized Sharpe ratios (Sharpe), while the third column indicates the corresponding clusters. An asterisk (*) denotes that the factor is included in the top B-C-IPCA model selected using the method by Chib et al. (2024). Cluster abbreviations are as follows: Beta = beta; FS = financial stability; FunMom = fundamental momentum; FunVol = fundamental volatility; GA = growth of assets; GE = growth of equity; Gr = growth; Illiq = illiquidity; Inv = investment; Mom = momentum; OE = operating efficiency; OI = operating illiquidity; Prof = profitability; PD = price delay; RV = return volatility; RetMom = return momentum; RD = R&D; Size = size; Turn = turnover; Val = value.

MFs	Train: 240 months		Train: 300 months		Train: 240 months		Train: 360 months	
	Sharpe	Clusters	Sharpe	Clusters	Sharpe	Clusters	Sharpe	Clusters
1	0.77	OI	0.82	OI	1.18	OI*	1.45	OE
2	0.55	RV*	0.61	Mom	0.69	FundMom*	1.19	Beta*
3	0.48	Size	0.56	FunVol	0.57	Inv	1.16	Prof*
4	0.44	GE*	0.55	Val	0.52	RV*	1.06	Size*
5	0.41	RetMom*	0.51	FunMom*	0.50	RetMom*	0.79	Mom*
6	0.33	OE	0.46	RD*	0.49	Size	0.71	FS*
7	0.28	GA	0.30	Size	0.45	RD	0.66	Gr*
8	0.24	FunMom*	0.15	Prof	0.34	Turn	0.49	Inv
9	0.17	Beta*	0.06	Illiq*	0.19	PD	0.34	Val*
10	0.03	SI	0.01	PD	0.18	FS	0.17	FunMom
11	0.03	ChgOI	0.93	Market*	0.85	Market*	0.03	RD
12	0.02	Inv					0.01	SI
13	0.61	Market*					1.04	Market*

Appendices

A Characteristics in IC and DC

Table A1: Characteristics in IC and DC

This table presents the characteristics of each IC and DC cluster. The first column outlines the economic meanings of the six clusters in the IC, while the second column describes the economic meanings of the thirteen clusters in the DC (see Figure 1 for cluster abbreviations). The final column lists the characteristics associated with each IC and DC cluster (see Table A.6 of Gu et al. (2020) for characteristic abbreviations). We use the same set of characteristics as in Gu et al. (2020), adopting their abbreviations accordingly.

IC	DC	characteristics
Mom	Mom	chmom, chtx, ear, indmom, mom12m, mom1m, mom36m, mom6m, nincr
	RV	baspread, idiovol, maxret, retvol
	S&I	dolvol, ill, mve, mve_ia, std_dolvol, zerotrade
TFs	PD	aeavol, pricedelay
	TO	beta, betasq, std_turn, turn
	Inv	cinvest, depr, pchcapx_ia, pchdepr
Inv		agr, chcscho, chinvt, egr, grcapx, grltnoa, invest, lgr
	Gr	chatoia, chpmia, ps, rsup, tb
Prof		cashpr, gma, lev, ms, operprof, roaq, roeq, roic
	Prof	bm, bm_ia, cashdebt, sp
Val		cfp, cfp_ia, dy, ep
	val	absacc, divi, divo
	Int	age, convind, herf, orgcap, rd_mve, rd_sale, realestate, roavol, salerec, secured, securedind, sin, stdacc, stdcf
Int		cash, currat, quick, salecash, tang
	OE	acc, pchsale_pchinvt, pchsale_pchrect, pchsaleinv, pctacc, saleinv
	Gr	chempia, hire, pchcurrat, pchgm_pchsale, pchquick, pchsale_pchxsga, rd, sgr

B Model Performance with Equal Weights

The empirical results suggest that the optimal clustering hyper-parameters are $\{m=19, knn=15, K=7\}$, with the resulting clustering denoted as DC^{EW} . This clustering yields seven distinct clusters, labeled as: Return momentum (RetMom), Size&illiquidity (S&I), Return volatility (RV) Turnover (TO), Operating illiquidity (OI), Intangibles (Int), and Short-run reversal (SR)¹⁰. A summary of the firm characteristics clustered in each cluster under both IC and DC^{EW} is provided in Table B1.

Consistent with Section 5.4 in the main text, we construct the DC^{EW} -IPCA model based on the seven clusters of firm characteristics. We then compare the performance of DC^{EW} -IPCA, IC-IPCA, and the standard IPCA by selecting a subset of factors using the ordered model selection and Bayesian model selection approaches, as described in Subsections 5.4.1 and 5.4.2. The comparison results are reported in Tables B2 and B3.

Similar to the value-weighted results presented in the main text, these tables demonstrate that even under an equal-weighted specification, the DC-IPCA model performs at least as well as the standard IPCA model in most cases. This indicates that our main findings are robust to the choice of weighting scheme. Overall, the use of equal weights still allows our approach to improve model interpretability without materially sacrificing performance.

¹⁰The SR cluster includes firm characteristics originally clustered in the Value (Val), Profitability (Prof), and Investment (Inv) clusters in the IC. However, among these characteristics, the estimated loading on characteristic 1-month momentum (mom1m) in Γ in equation (2) is substantially higher than for the other characteristics. Since Γ reflects the sensitivity of firm characteristics to risk exposures (β in equation (2)), this suggests that the risk exposure in this cluster is primarily driven by short-run reversal. Accordingly, we label the cluster as SR.

Table B1: Characteristics in IC and DC^{EW}

This table presents the characteristics in each IC and DC^{EW} cluster. The first column outlines the economic meanings of the six clusters in the IC, while the second column describes the economic meanings of the seven clusters in the DC^{EW} . Cluster abbreviations are as follows: Return momentum(RetMom), Size&illiquidity(S&I), Return volatility(RV) Turnover(TO), Operating illiquidity(OI), Intangibles(Int), Short-run reversal(SR), Trading frictions(TFs), Value(Val), Profitability(Prof), Investment(Inv), Momentum(Mom). The final column lists the characteristics associated with each IC and DC^{EW} cluster (see Table A.6 of [Gu et al. \(2020\)](#) for characteristic abbreviations). We use the same set of characteristics as in [Gu et al. \(2020\)](#), adopting their abbreviations accordingly.

IC	DC	characteristics
TFs	RV	baspread, idiovol, maxret, retvol
	S&I	dolvol, ill, mve, mve_ia, std_dolvol, zerotrade
	TO	aeavol, beta, betasq, pricedelay, std_turn, turn
Int	OI	cash, currat, quick, salecash, tang
	Int	age, convind, divo, herf, orgcap, rd, rd_mve, rd_sale, realestate, saleinv, salerec, secured, securedind, sin, stdacc, stdcf
	SR	
Val		bm, bm_ia, cashdebt, cfp, cfp_ia, dy, ep, sp
Prof		cashpr, chatoia, chpmia, gma, lev, ms, operprof, ps, roaq, roeq, roic, rsup, tb
Inv		agr, chcscho, chin, cinvest, depr, egr, grcapx, grltnoa, invest, lgr, pchcapx_ia, pchdepr
Mom		chtx, ear, mom1m, mom36m, nincr
	RetMom	chmom, indmom, mom12m, mom6m

Table B2: Mean-variance efficiency of the Ordered C-IPCA, IPCA and Ordered IPCA models under an equal-weighted scheme

This table reports the out-of-sample Sharpe ratios of tangency portfolios using an equal-weighted scheme for Ordered C-IPCA models and two benchmarks: (i) the IPCA model with the same number of factors, and (ii) the Ordered IPCA models. The J factor Ordered C-IPCA model consists of the first $J - 1$ factors from the corresponding C-IPCA specification and the market factor, with factors ordered by their Sharpe ratios in the training sample (1985:01–1999:12). The Ordered IPCA benchmark for a given Ordered C-IPCA model includes the first J factors from an IPCA model. Specifically, the Ordered IPCA model paired with Ordered IC-IPCA (Panel A) is based on an IPCA model with 7 factors (Ordered IPCA7), while the benchmark for Ordered DC-IPCA (Panel B) is based on an IPCA model with 8 factors (Ordered IPCA8). Tangency portfolios are constructed entirely out-of-sample by estimating the mean and covariance matrix of model factors using data up to time t and computing the portfolio return at $t + 1$. The first column lists the number of factors (J). Columns two through four report the annualized Sharpe ratios for Ordered C-IPCA, IPCA, and Ordered IPCA models, respectively. The final column identifies the cluster associated with the newly added factor in the Ordered C-IPCA model. (See Table B1 for cluster abbreviations; “Mkt” refers to the zero-cost market factor.) The sample period spans January 2000 through December 2021.

Panel A. IC				
J	Ordered IC-IPCA	IPCA	Ordered IPCA7	Econ. Interp.
1	0.37	0.30	0.46	Mkt
2	0.51	1.05	0.70	Inv
3	0.75	1.23	0.66	Prof
4	1.18	1.25	0.65	Mom
5	1.25	1.67	0.70	Val
6	1.55	1.56	0.72	Int
7	1.50	1.57	1.57	TFs
Panel B. DC^{EW}				
J	Ordered DC^{EW} -IPCA	IPCA	Ordered IPCA8	Econ. Interp.
1	0.19	0.30	0.62	Mkt
2	0.85	1.05	0.54	RetMom
3	0.75	1.23	0.72	Int
4	1.54	1.25	0.77	SR
5	1.84	1.67	0.81	RV
6	1.87	1.56	1.79	OI
7	1.86	1.57	1.79	TO
8	1.90	1.78	1.78	S&I

Table B3: Mean-variance efficiency of the Bayesian C-IPCA and Bayesian IPCA models under an equal-weighted scheme

This table reports the out-of-sample Sharpe ratios of tangency portfolios using the equal-weighted scheme for the top 10 Bayesian DC^{EW} -IPCA models (Panel A) and Bayesian IC-IPCA models (Panel B) with the highest posterior probabilities. In each Panel, we also show the two benchmarks for the Bayesian C-IPCA models: Bayesian IPCA models and standard IPCA models with the same number of factors, estimated following Section 5.4.2. For each model, we show Sharpe ratio of their tangency portfolios (Sharpe) and the number of factors (# factors). All tangency portfolios are constructed on a purely out-of-sample basis, using all observations up to time t to estimate the mean and covariance matrix, with portfolio returns evaluated at $t + 1$. The sample period spans January 2000 through December 2021.

		Top1	2	3	4	5	6	7	8	9	10
Panel A. DC^{EW} -IPCA											
Bayesian DC^{EW} -IPCA	Sharpe	1.76	1.78	1.76	1.73	1.84	1.78	1.87	1.86	1.79	1.82
	# factors	4.00	5.00	6.00	5.00	5.00	5.00	6.00	7.00	7.00	6.00
Bayesian IPCA8	Sharpe	0.81	0.79	1.79	0.77	0.81	1.78	1.81	0.79	0.77	1.79
	# factors	5.00	6.00	6.00	4.00	6.00	7.00	5.00	6.00	4.00	7.00
IPCA	Sharpe	1.25	1.67	1.56	1.67	1.67	1.67	1.56	1.57	1.57	1.56
	# factors	4.00	5.00	6.00	5.00	5.00	5.00	6.00	7.00	7.00	6.00
Panel B. IC-IPCA											
Bayesian IC-IPCA	Sharpe	0.85	1.10	0.88	1.12	0.93	1.09	0.84	1.21	0.33	0.62
	# factors	4.00	5.00	3.00	4.00	5.00	6.00	5.00	6.00	4.00	4.00
Bayesian IPCA7	Sharpe	0.68	0.65	0.72	0.70	1.57	1.51	1.57	1.51	0.72	0.75
	# factors	5.00	4.00	6.00	5.00	6.00	5.00	7.00	6.00	4.00	5.00
IPCA	Sharpe	1.25	1.67	1.23	1.25	1.67	1.56	1.67	1.56	1.25	1.25
	# factors	4.00	5.00	3.00	4.00	5.00	6.00	5.00	6.00	4.00	4.00

C Rationality of Chameleon

Section 3.3.3 argues that Chameleon is robust to data noise because of the merging rule in equation (19). This section details this argument.

We first recall that firm characteristics within the same cluster are assumed to be noisy measurements of a common latent risk exposure. In the absence of measurement errors, each characteristic would perfectly measure the latent exposure, resulting in a correlation of 1 between characteristics within the same cluster, thus maximizing the similarity s_{ij} in equation (17). However, as measurement errors increase, the observed characteristics deviate more from the latent exposure, reducing their correlation and lowering their similarity.

The inclusion of intra-cluster similarity, $INTRA(C_i)$, in the merging rule (19) allows the algorithm to detect clusters with high measurement errors and prevents them from being erroneously split into multiple sub-clusters. For illustration, consider four clusters of firm characteristics. The first two clusters, shown in panels (a) and (b) of Figure C1 (copied from Karypis et al. (1999)), exhibit low measurement errors and thus high intra-cluster similarity (depicted by small intra-cluster distances). In contrast, the other two clusters, shown in panels (c) and (d), have high measurement errors and low intra-cluster similarity (depicted by large intra-cluster distances).

The task is to determine whether to merge clusters (a) and (b), or (c) and (d). Intuitively, merging (c) and (d) seems more reasonable, as their intra-cluster distance appears comparable to their inter-cluster distance, indicating high homogeneity between them. If the algorithm only considers inter-cluster similarity (or distance), it would incorrectly merge (a) and (b) due to their smaller inter-cluster distance. However, by incorporating both intra-cluster and inter-cluster distances, as in the Chameleon algorithm, the algorithm correctly merges (c) and (d), aligning with intuition. Specifically, Chameleon avoids merging (a) and (b) because such a merger would increase the intra-cluster distance and reduce the Relative Inter-Cluster Similarity (RIS) in equation (19). Thus, the inclusion of intra-cluster similarity ensures that high-noise sub-clusters, such as (c) and (d), are merged into a single high-noise cluster, preventing the erroneous splitting of a high-noise cluster into multiple sub-clusters.

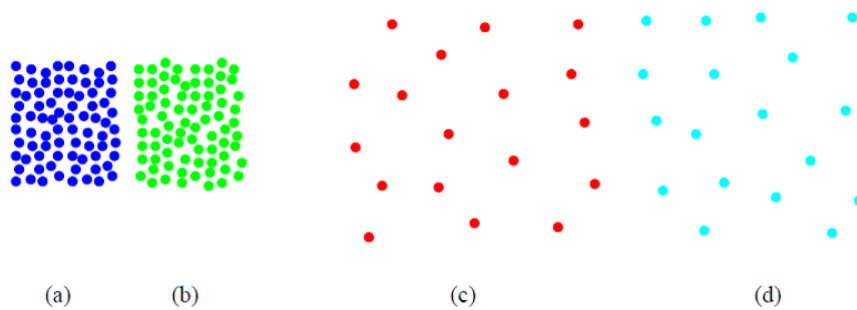


Figure C1: Clustering with Noise

D Bayesian model selection method

Bayesian Model Framework. We adopt the Bayesian framework developed by Chib et al. (2020) for model comparison across different factor sets. Specifically, consider a C-IPCA model with J factors ($J = 7$ for IC-IPCA and $J = 13$ for DC-IPCA). The J factors results in $L = 2^J - 1$ candidate models, each corresponding to a subset of the J factors. As shown in Chib et al. (2020), this framework enables the estimation of the posterior probability for each potential model, based on observed data D . We select the top 10 models with the highest posterior probabilities. The technical details of this method is available in Appendix D

The basic idea of this framework is outlined below, with further details available in Chib et al. (2020).

Formally, let $M_1, M_2, \dots, M_L$ represent the L candidate models, where each model M_l consists of a subset of factors from the full set of C-IPCA factors. We begin by assigning each model an equal prior probability, reflecting a non-informative prior over the model space:

$$\Pr(M_l) = \frac{1}{L}, \quad \forall l = 1, \dots, L.$$

This assignment implies that, prior to observing the data, all models are equally likely.

Posterior Probability Calculation. Upon observing the data D , we update the prior using Bayes' theorem to compute the posterior probability of each model:

$$\Pr(M_l|D) = \frac{\Pr(M_l) \Pr(D|M_l)}{\sum_{i=1}^L \Pr(M_i) \Pr(D|M_i)} \propto \Pr(D|M_l),$$

where $\Pr(M_l)$ is the prior probability of model M_l , and $\Pr(D|M_l)$ is the likelihood of the data given that model M_l is correct.

Likelihood of the Data. The likelihood $\Pr(D|M_l)$ represents the probability of observing the data D given that model M_l is the true model. The data D consists of two components: the C-IPCA factors and the test-assets returns. As noted in Chib et al. (2020), the relative posterior probability between models with various subsets of factors is invariant to the choice of test-assets returns. For simplicity, we assume an empty set for the test-assets returns in this context.

For each model M_l , let f_l denote the set of C-IPCA factors included in the model, and f_l^* the set of factors excluded from the model. The factors f_l included in the model are assumed to be linear combinations of the corresponding characteristics, with an intercept term α_l and an error term ε_l :

$$f_l = \alpha_l + \varepsilon_l,$$

where $\varepsilon_l \sim N(0, \Sigma_l)$ which can be estimated from the data. For the factors not included in model M_l , we write

$$f_l^* = \alpha_l^* + \beta_l^* f_l + \varepsilon_l^*, \quad (29)$$

where $\varepsilon_l^* \sim N(0, \Sigma_l^*)$. Following Chib et al. (2020), if the factors f_l are mean-variance efficient, then $\alpha_l^* = 0$. The relative posterior probability across all models that include a subset of factors is proportional to the probability of explaining f_l^* given the factors f_l included in the model.

The prior distribution for the intercept α_l is assumed to be normal:

$$\alpha_l | \Sigma_l \sim N(\alpha_{l0}, k_l \Sigma_l),$$

where α_{l0} will be estimated from the data, and k_l is a scaling factor related to the maximum achievable squared Sharpe ratio in the market:

$$k_l = \frac{\text{Sh}_{\max}^2}{J},$$

where J is the number of factors in model M_l , and $\text{Sh}_{\max}^2$ is the maximum achievable squared Sharpe ratio.

Estimation of Parameters. To estimate the parameters α_{l0} , Σ_l , and k_l , we split the training sample into two sub-samples. The first sub-sample consists of the first $tr \times T$ months, where T is the total length of the training sample and tr is the proportion used for prior estimation. This sub-sample is used to estimate the prior parameters α_{l0} and Σ_l , with $\hat{\alpha}_l$ and $\hat{\Sigma}_l$ representing the mean and standard deviation of the factors. The second sub-sample consists of the remaining months, used to compute the posterior probability $\Pr(M_l|D)$.

Model Selection. For each model M_l , based on a subset of C-IPCA factors, we calculate the posterior probability $\Pr(M_l|D)$ following Chib et al. (2020). The 10 models with the highest posterior probabilities are selected to form the Bayesian C-IPCA model.